

Electron Transport Chain (ETC) & Oxidative Phosphorylation

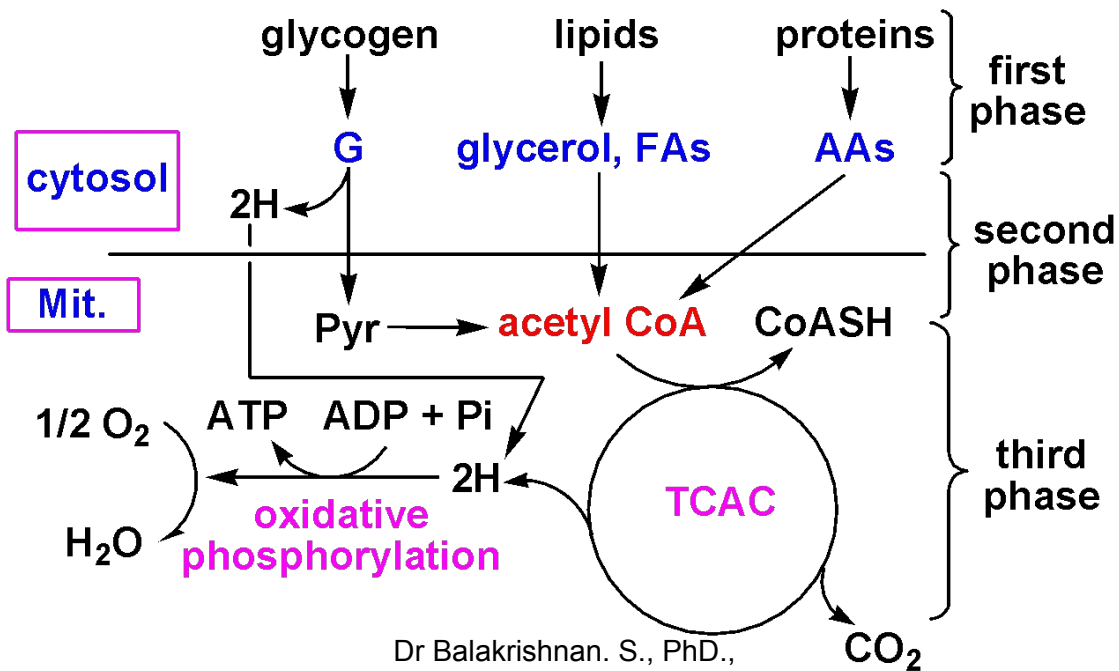
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Objectives:

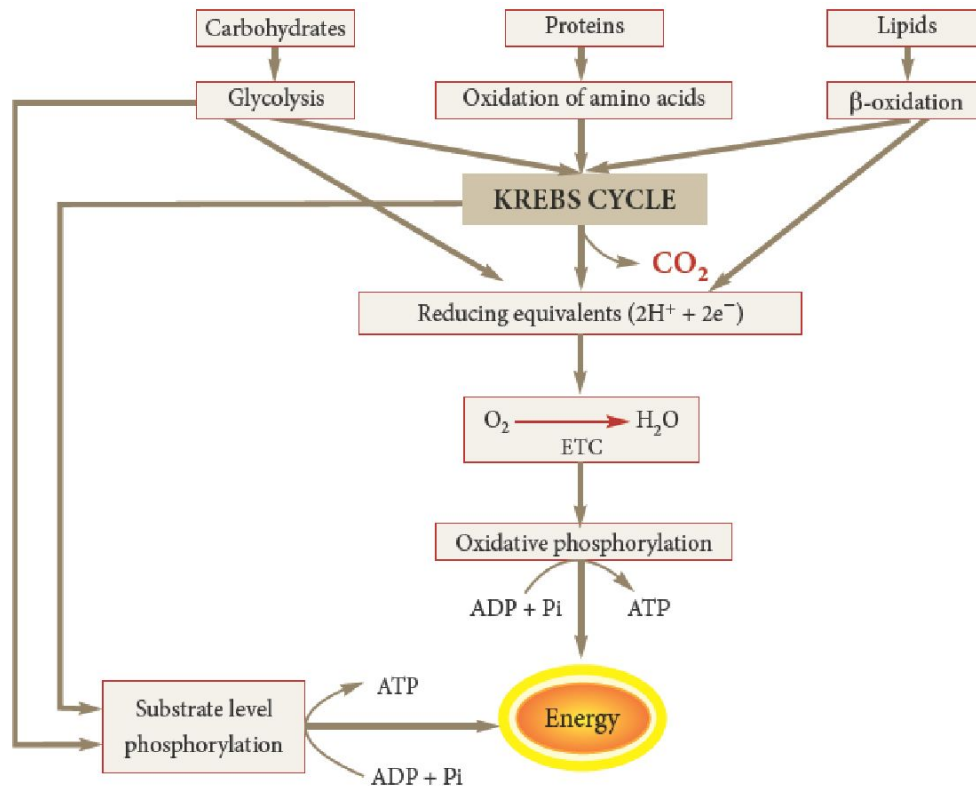
- At the end of this chapter, it is expected to;
 - Understand the concepts of Biological oxidation and Respiratory Chain (ETC)
 - Appreciate the importance of ETC
 - Describe the Structure of ETC
 - Describe the functioning of the ETC
 - Understand the concepts of Oxidative Phosphorylation
 - Appreciate the importance of Oxidative Phosphorylation
 - Describe the coupling of ETC and Oxidative Phosphorylation
 - Explain the Regulation of ETC and oxidative phosphorylation
 - Identify the Inhibitors of ETC
 - Explain the deficiencies of ETC and oxidative phosphorylation

Biological oxidation

- The process in which substances (carbohydrate, Lipid, AAs) are oxidized in living organism.

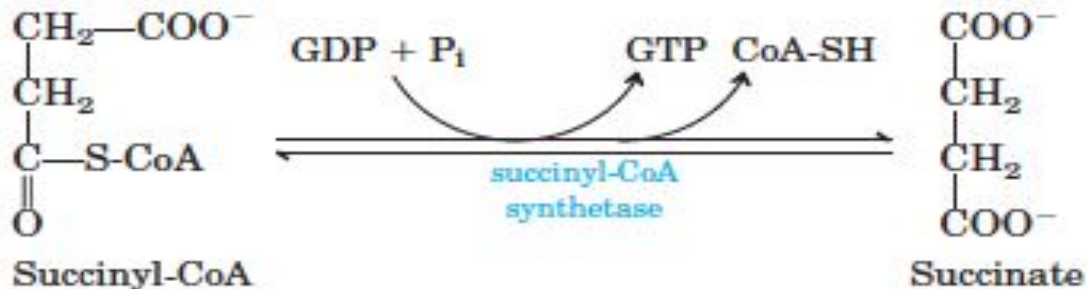


Pathways linking the breakdown of organic nutrients to the generation of ATP

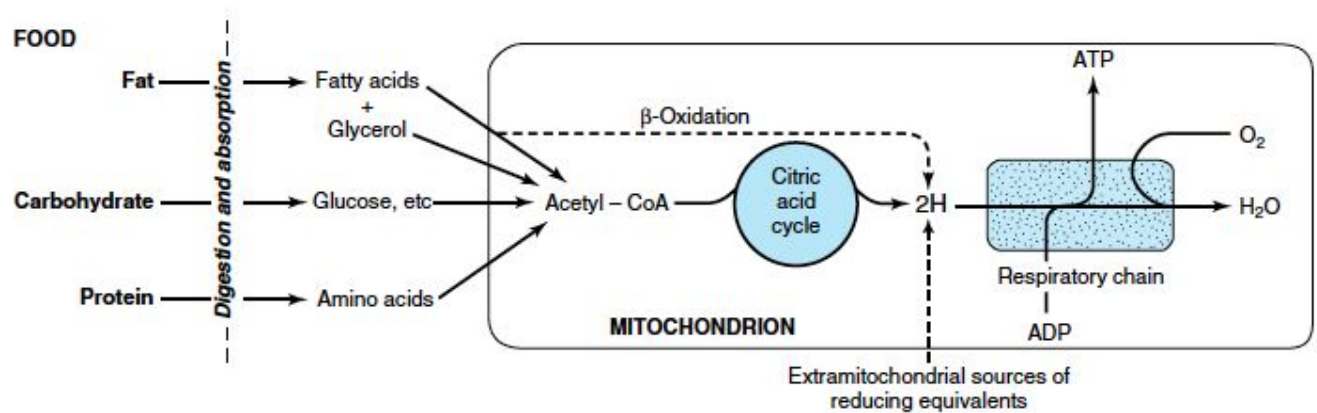


Substrate level phosphorylation:

- Phosphorylation of ADP or GDP to ATP or GTP coupled to the dehydrogenation of an organic substrate.
- Example: Conversion of succinyl-CoA to succinate by Succinyl-CoA Synthetase in TCA cycle.



Role of the respiratory chain of mitochondria in the conversion of food energy to ATP.



- Oxidation of the major foodstuffs leads to the generation of reducing equivalents ($2H$) that are collected by the respiratory chain for oxidation and coupled generation of ATP

Nature of biological oxidation :

- 1. 37°C, pH 7.4, enzymatic reaction.**
- 2. Energy released gradually.**
- 3. Formation of H₂O.**
- 4. Formation of CO₂ by decarboxylation.**

ENZYMES INVOLVED IN BIOLOGICAL OXIDATION

The crucial enzymes of biological oxidation belong to the group ***oxidoreductases***. They are classified into four categories.

1. Oxidases

- Cytochrome oxidase, Flavoproteins

2. Dehydrogenases

- Cytochromes, $\text{NAD}^+/\text{NADP}^+$ dependent dehydrogenases

3. Hydroperoxidases

- Peroxidase, catalase

4. Oxygenases

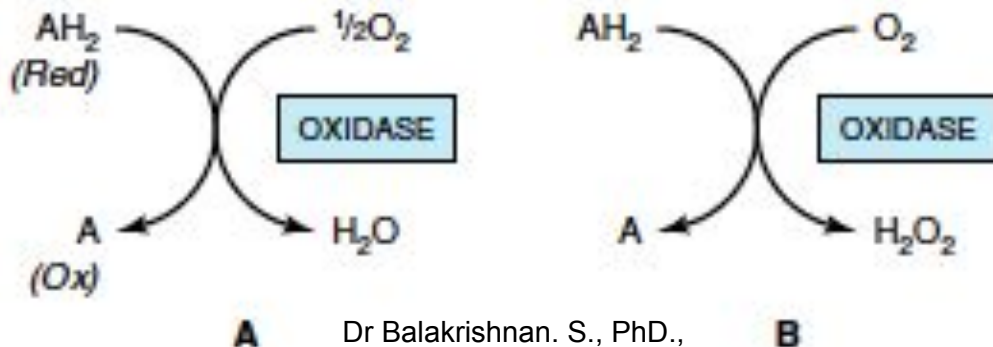
- Monooxygenases, Dioxygenases

OXIDASES:

These enzymes catalyse the removal of hydrogen from a substrate and hand it over to oxygen to form water.

- Cytochrome oxidase
- Flavoproteins

Oxidation of a metabolite catalyzed by an oxidase:
(A) forming H_2O ,
(B) forming H_2O_2 .



DEHYDROGENASES:

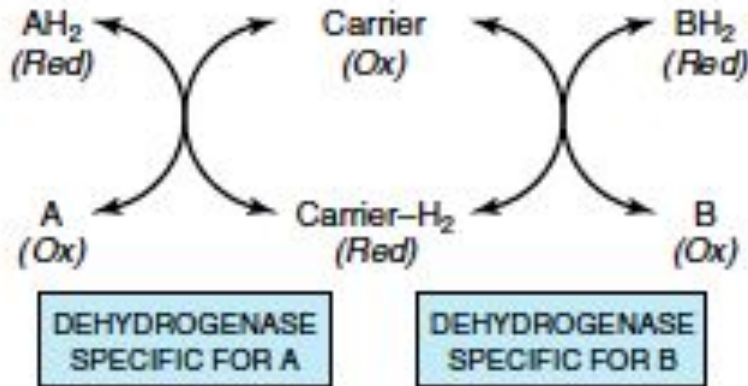
These enzymes transfer hydrogen from one substrate to another in a coupled oxidation–reduction reaction.

a) Cytochromes:

b) Dehydrogenases dependent on niacin coenzymes:

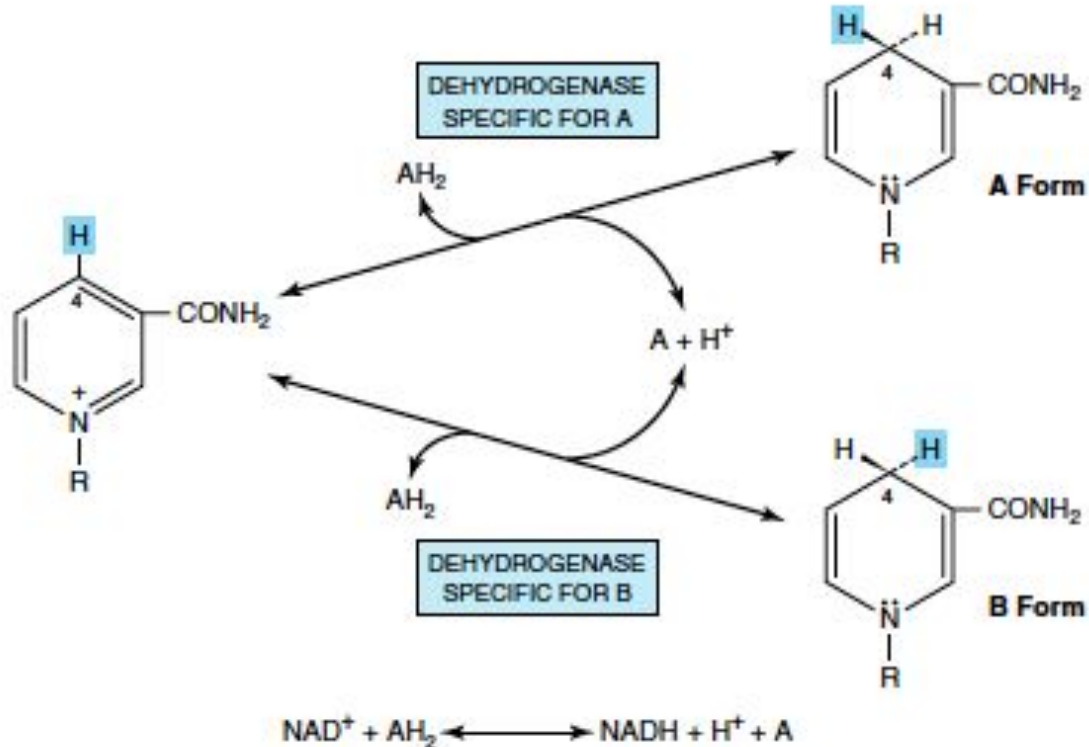
NAD⁺ dependent dehydrogenases

NADP⁺ dependent dehydrogenases



(Oxidation of a metabolite catalyzed by coupled dehydrogenases)

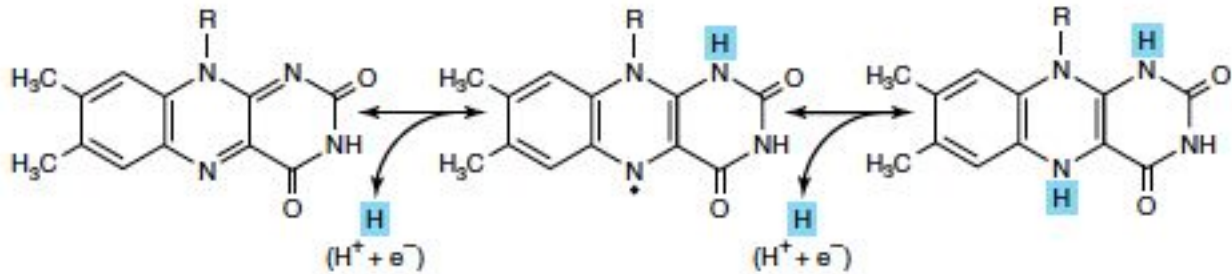
Mechanism of oxidation and reduction of nicotinamide coenzymes:



c) Dehydrogenases dependent on flavin coenzymes:

FMN-dependent dehydrogenases

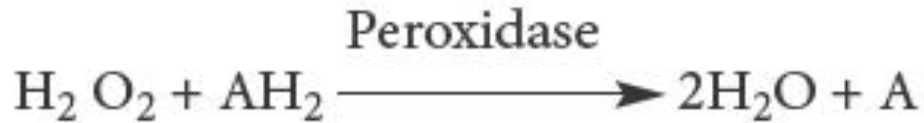
FAD dependent dehydrogenases



(Oxidoreduction of isoalloxazine ring in flavin nucleotides via a semiquinone (free radical) intermediate (center)).

HYDROPEROXIDASES:

Peroxidase



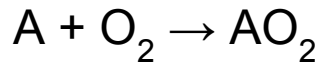
Catalase



OXYGENASES:

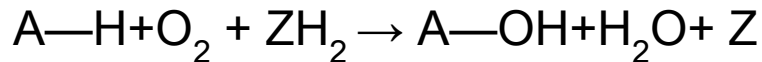
- Dioxygenases

- Dioxygenases incorporate both atoms of molecular oxygen into the substrate



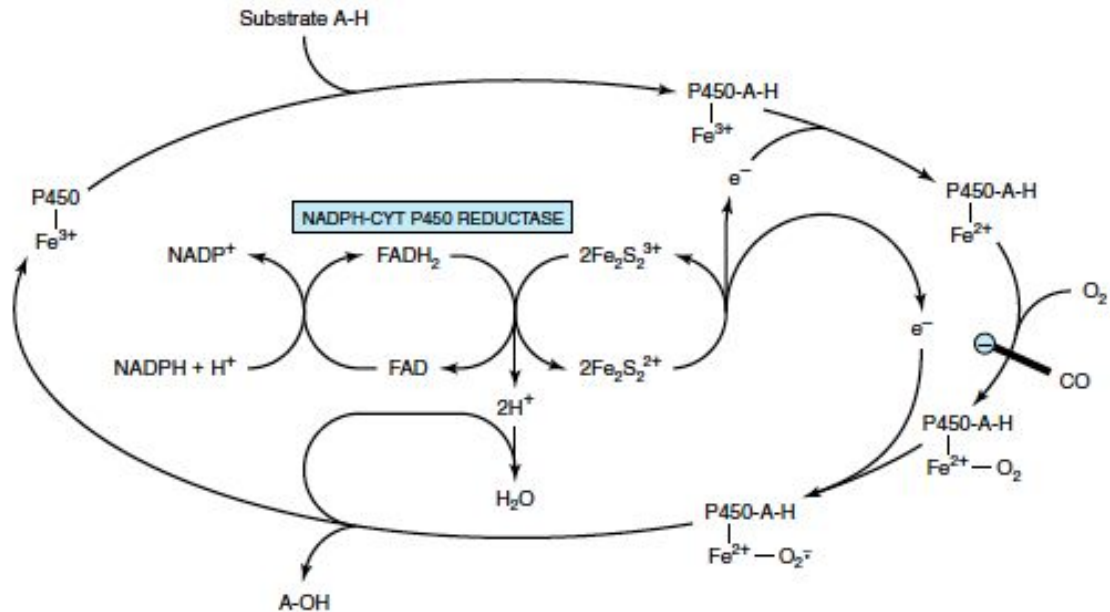
- Monooxygenases

- Monooxygenases (Mixed-Function Oxidases, Hydroxylases) incorporate only one atom of molecular oxygen into the substrate



Cytochromes P450:

- Cytochromes P450 are Monooxygenases
- Important for the Detoxification of Many Drugs & for the Hydroxylation of Steroids



(Cytochrome P450 hydroxylase cycle in microsomes)

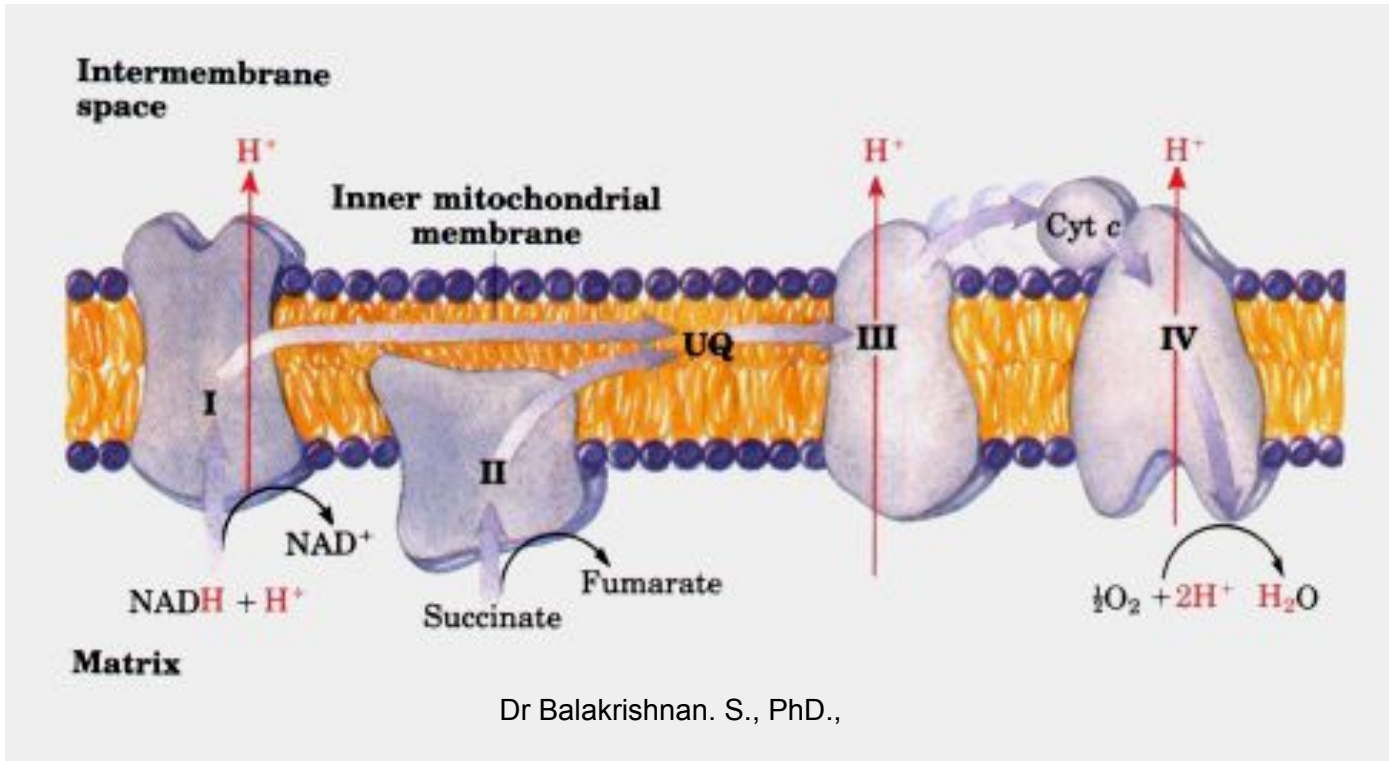
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Respiratory Chain/ Electron Transport Chain (ETC)

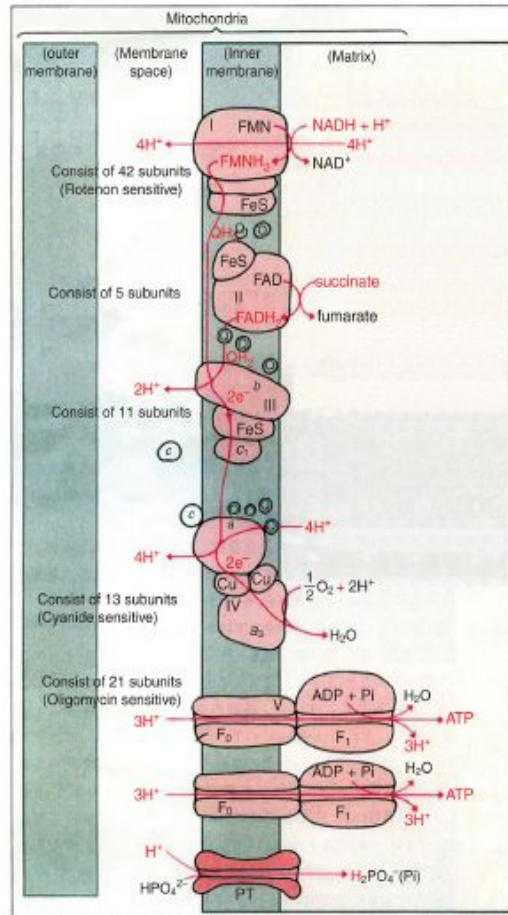
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Respiratory Chain/ ETC

- A chain of complexes in the mitochondria consists of a number of redox carriers for transferring hydrogens removed from the substrate to oxygen to form water.



Electron Transport Chain



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Composition of Respiratory Chain Complexes

Complexes	Name	Prosthetic Groups
Complex I	NADH-CoQ Reductase	FMN, Fe-S
Complex II	Succinate-CoQ Reductase	FAD, Fe-S
Complex III	CoQ-Cyt c Reductase	iron-porphyrine Fe-S
Complex IV	Cytochrome Oxidase	iron-porphyrine Cu

Composition of ETC

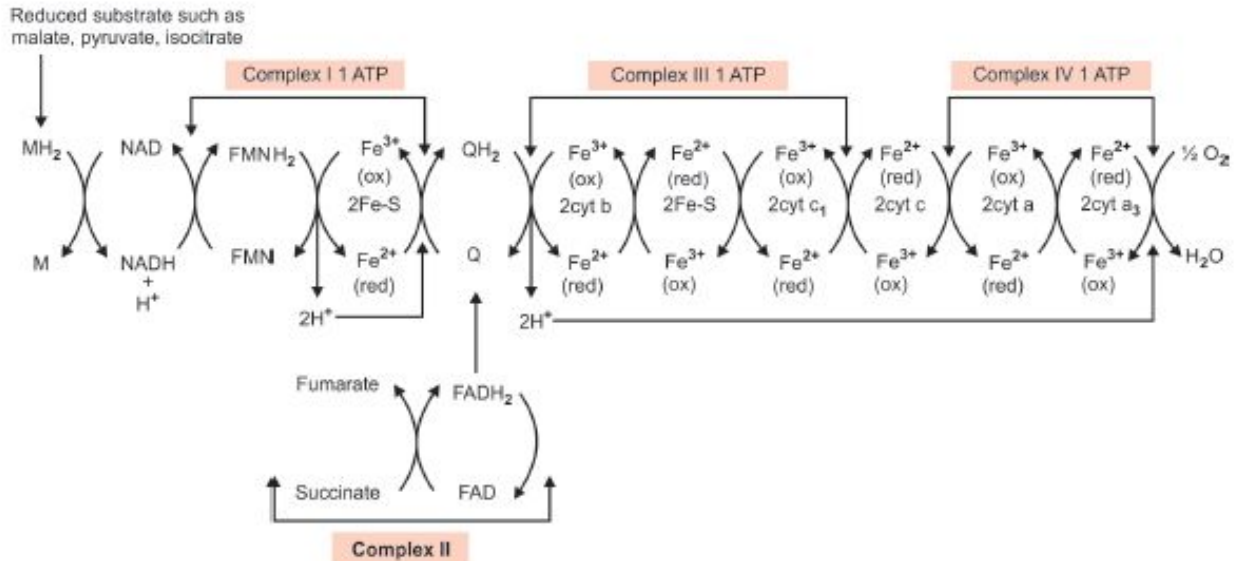
- Hydrogen carrier :
 - NAD^+
 - FMN
 - FAD
 - CoQ
- Electron carrier :
 - Fe-S
 - Cyt

Components of the mitochondrial electron-transport chain

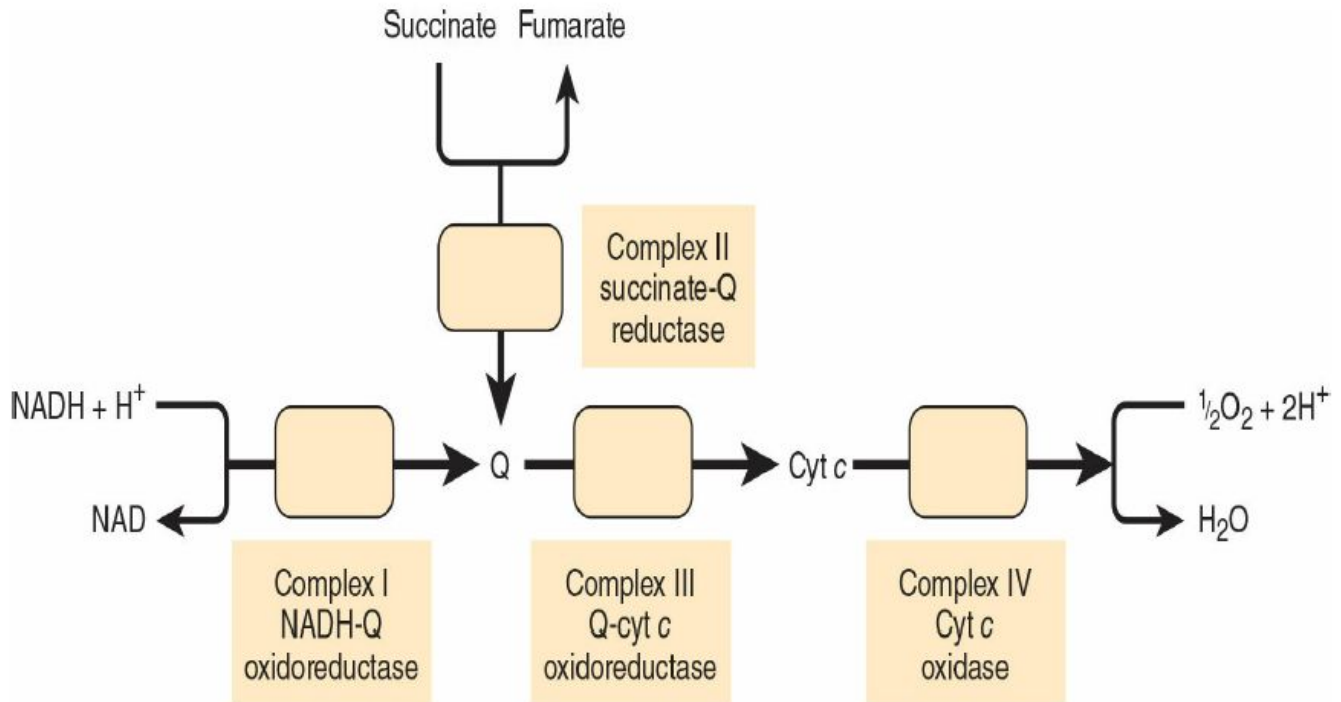
Enzyme complex	Mass (kd)	Subunits	Prosthetic group	OXIDANT OR REDUCTANT		
				Matrix side	Membrane core	Cytoplasmic side
NADH-Q oxidoreductase	>900	46	FMN Fe-S	NADH	Q	
Succinate-Q reductase	140	4	FAD Fe-S	Succinate	Q	
Q-cytochrome c oxidoreductase	250	11	Heme b_H Heme b_L Heme c_1 Fe-S		Q	Cytochrome c
Cytochrome c oxidase	160	13	Heme a Heme a_3 Cu _A and Cu _B			Cytochrome c

Sources: J. W. DePierre and L. Ernster. *Annu. Rev. Biochem.* 46(1977):215; Y. Hatefi. *Annu. Rev. Biochem.* 54(1985):1015; and J. E. Walker. *Q. Rev. Biophys.* 25(1992):253.

Electron transport chain

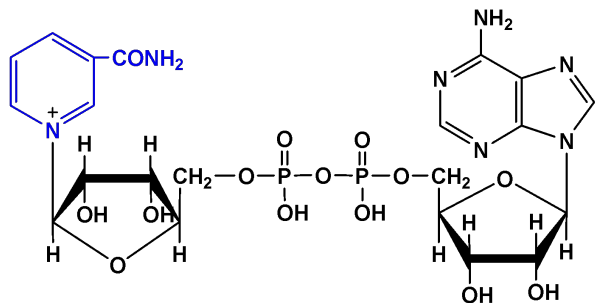


Overview of electron flow through the respiratory chain

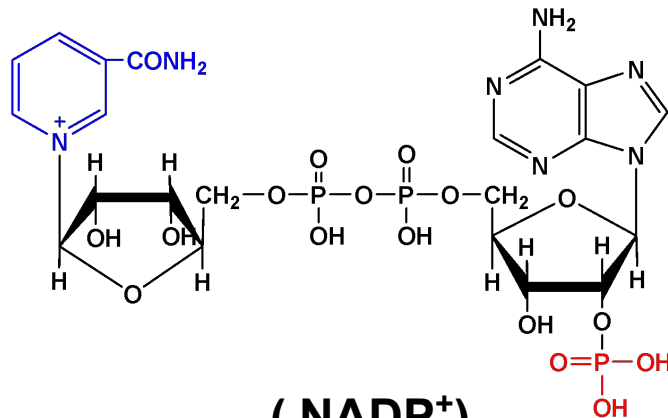


1. Nicotinamide coenzymes:

- NAD^+/NADH (Nicotinamide Adenine Dinucleotide, Co I)
- $\text{NADP}^+/\text{NADPH}$ (Nicotinamide Adenine Dinucleotide Phosphate, Co II)
- The nicotinamide is the vitamin B3.

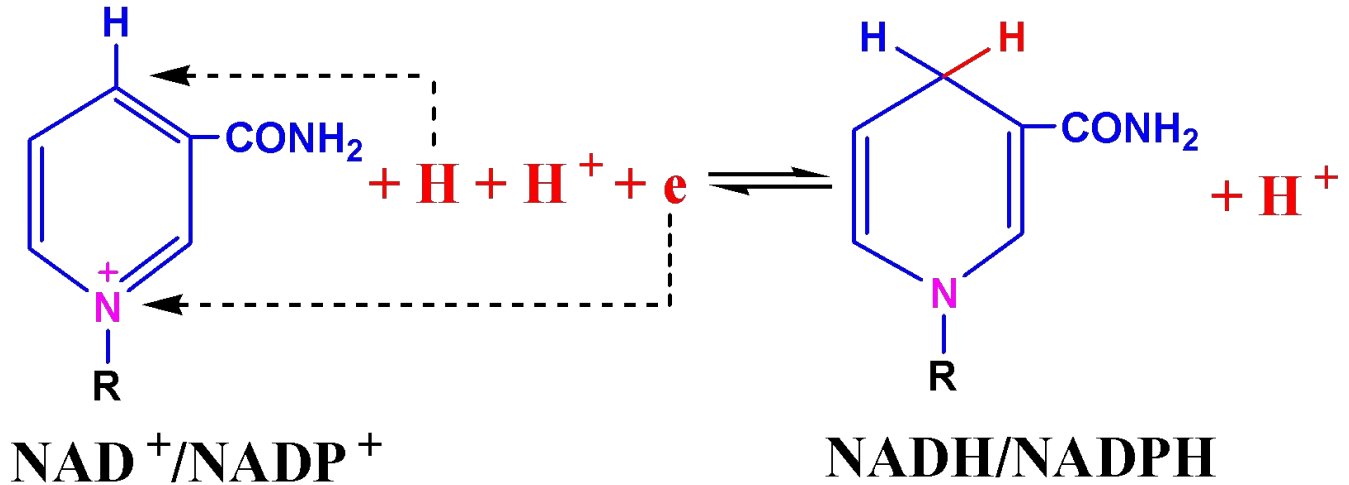


(NAD^+)



(NADP^+)

Oxidation/Reduction of NAD^+/NADH & $\text{NADP}^+/\text{NADPH}$

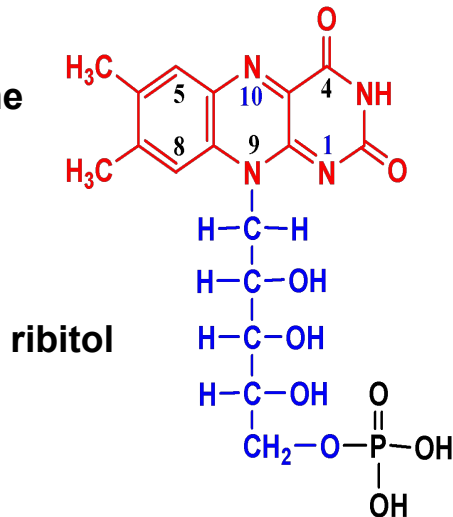


- ◆ NAD^+ is a coenzyme, that reversibly binds to enzymes.

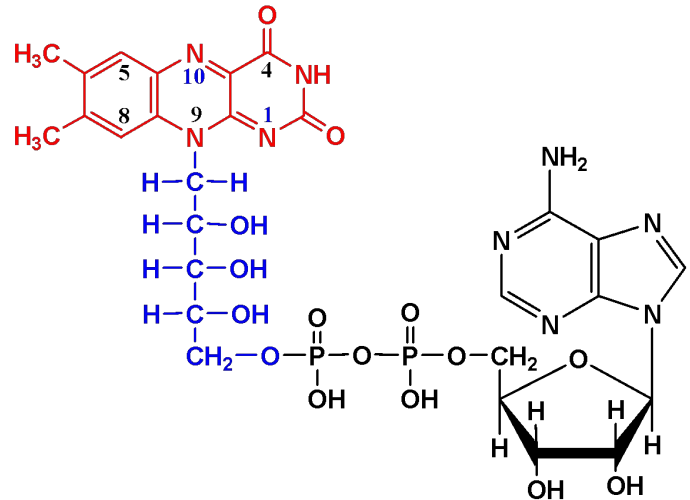
2. Flavin prosthetic groups:

- FMN: Flavin Mononucleotide
- FAD: Flavin Adenine Dinucleotide
- They contain the riboflavin (Vit B₂).

Isoalloxazine



(FMN)

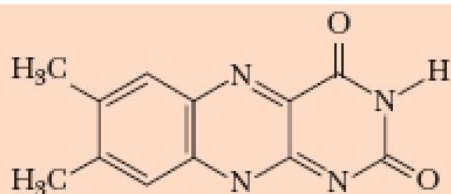


(FAD)

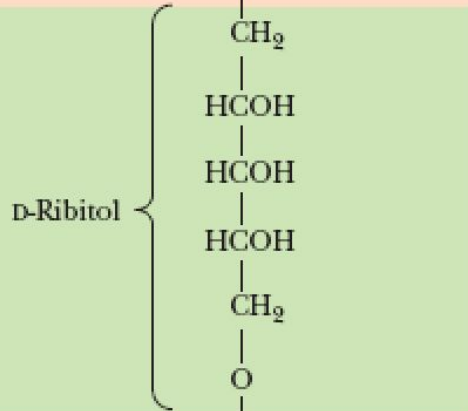
Flavin adenine dinucleotide, FAD

Flavin mononucleotide, FMN

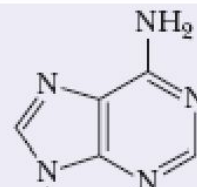
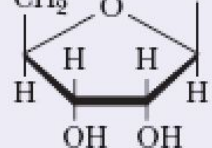
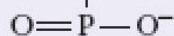
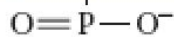
Riboflavin



Isoalloxazine

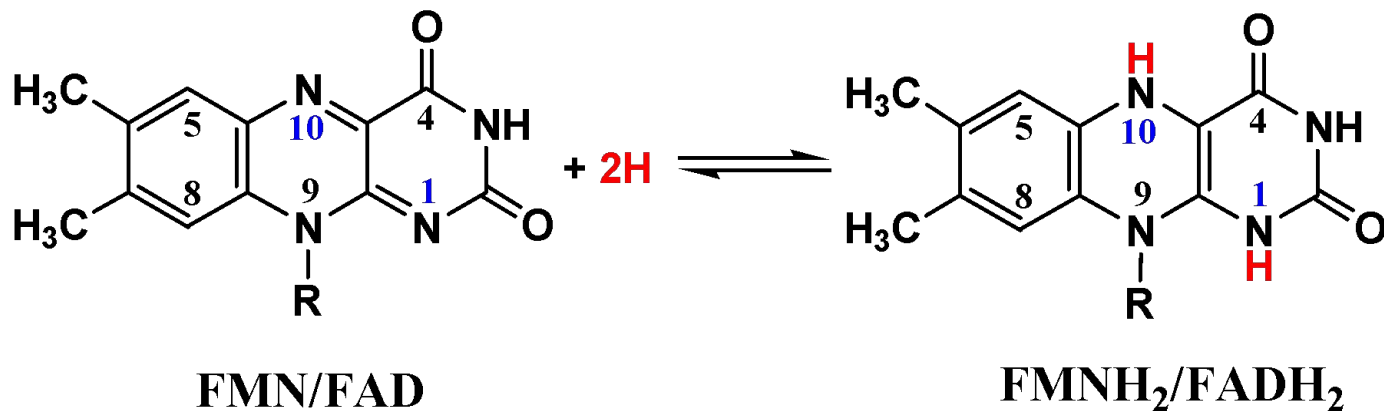


D-Ribitol



AMP

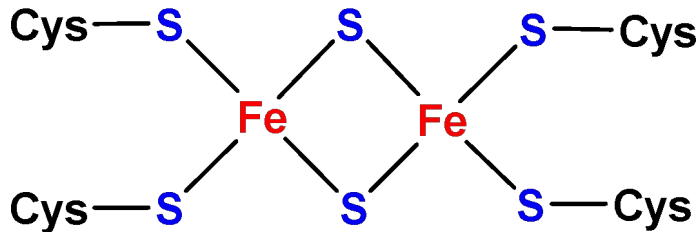
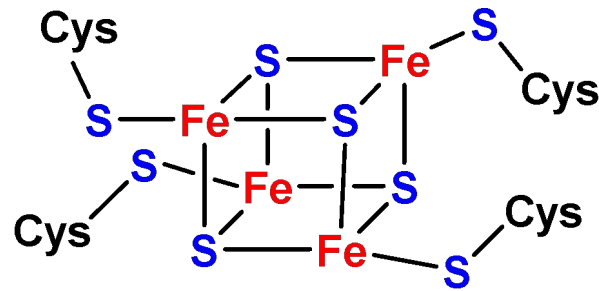
Oxidation/Reduction of FMN/FMNH₂ & FAD/FADH₂



- ◆ FAD is a prosthetic group, that remains tightly bound at the active site of an enzyme.

3. Fe-S:

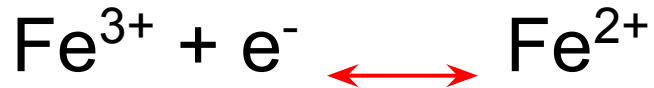
- Iron-sulfur centers (Iron-sulfur protein, Fe-S)
- Are prosthetic groups containing 2, 3, 4 or 8 nonheme iron atoms complexed to elemental and cysteine S.



(Iron-sulfur centers)

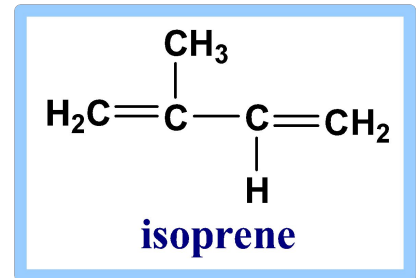
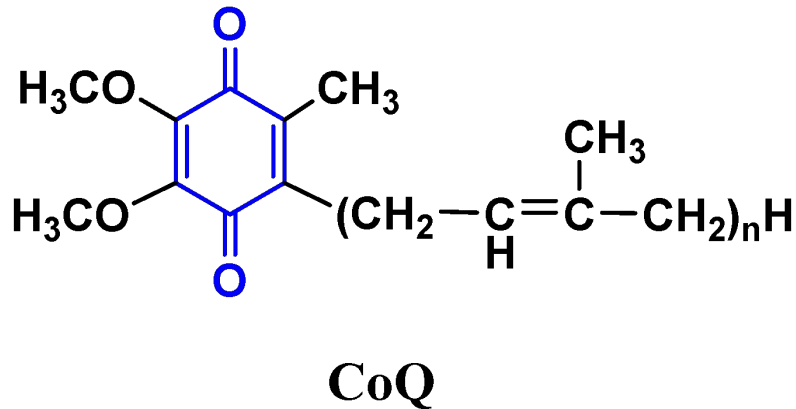
Oxidation/Reduction of Fe-S

- Iron-sulfur centers transfer only one electron.

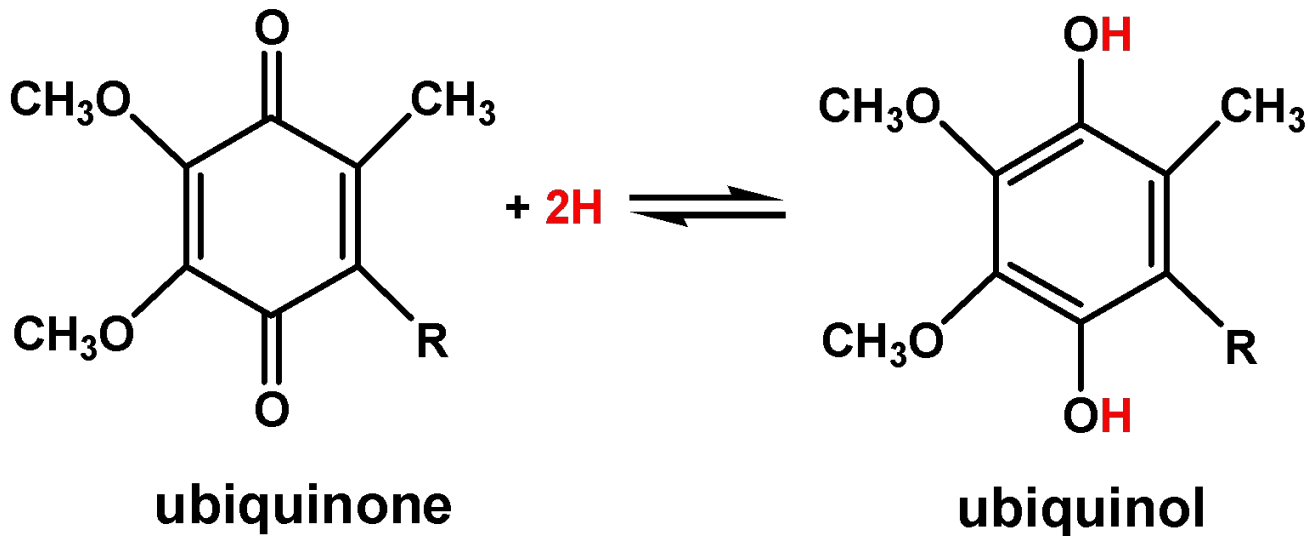


4. CoQ:

- Coenzyme Q (CoQ, ubiquinone) is very hydrophobic. It dissolves in the membrane.
- Coenzyme Q functions as a mobile e^- carrier within the mitochondrial inner membrane.

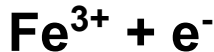


Oxidation/Reduction of CoQ

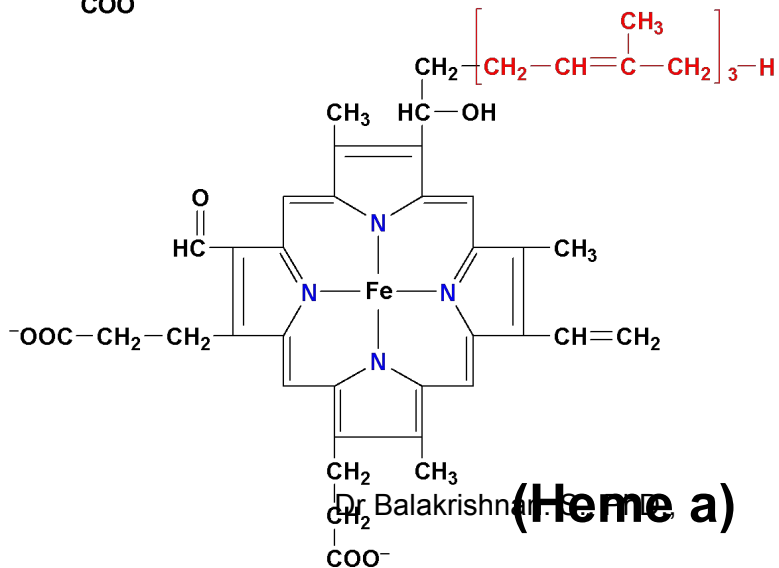
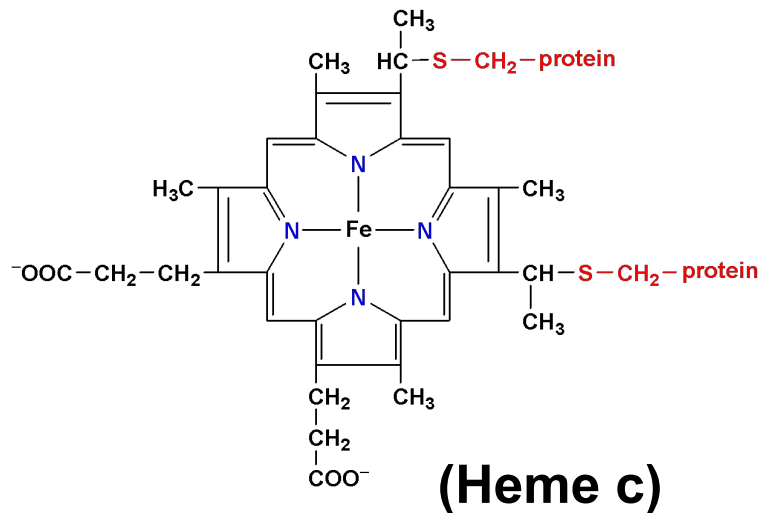
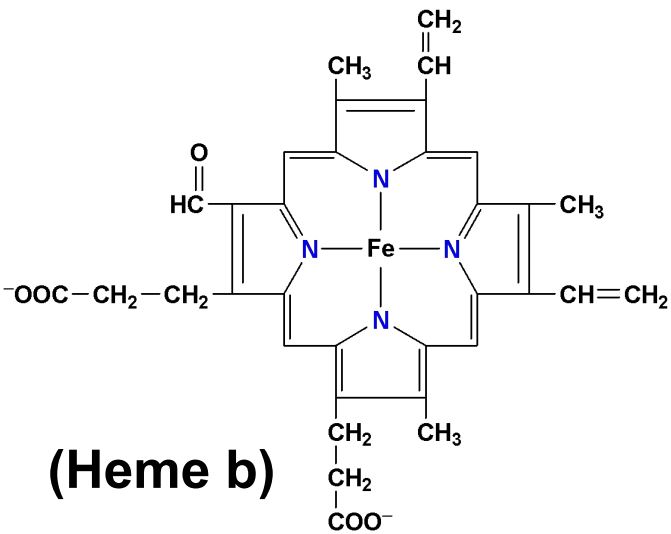


5. Cytochromes:

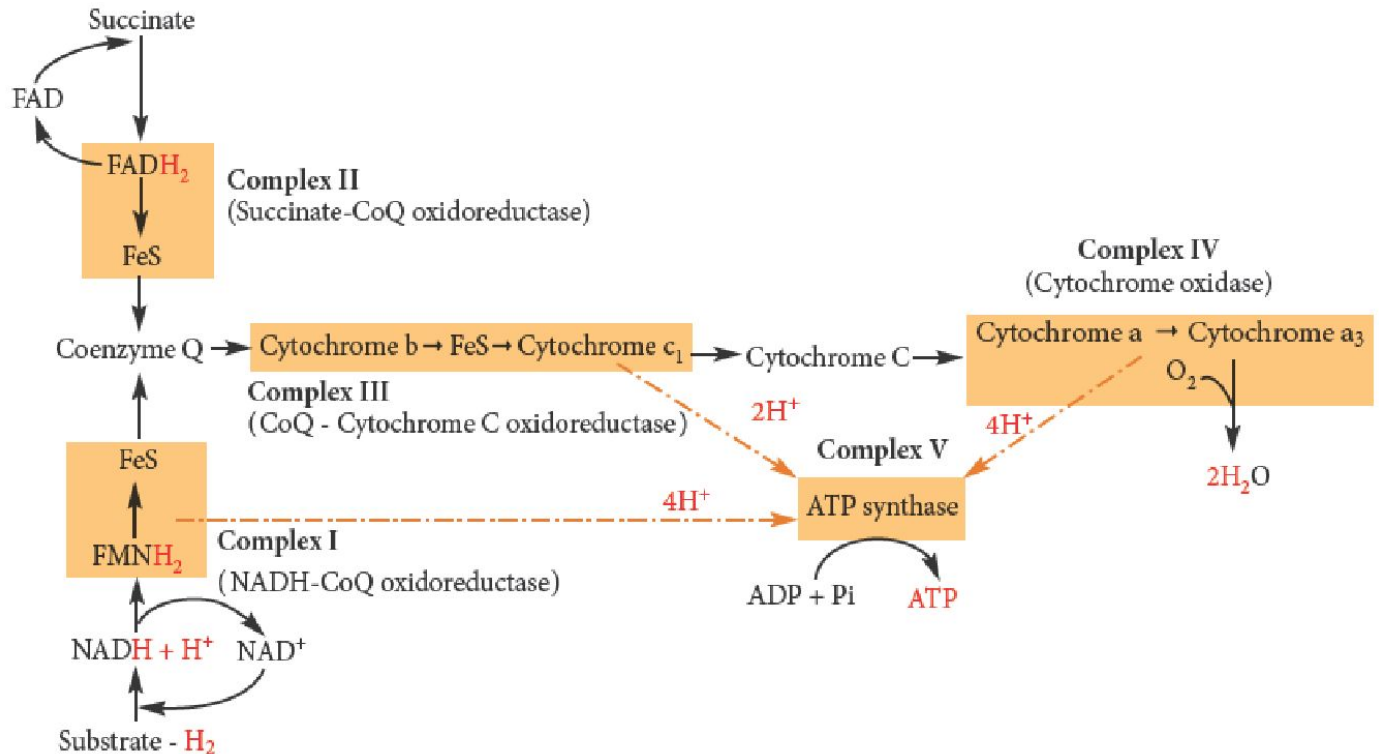
- Cytochromes (Cyt) are proteins with heme prosthetic groups. They absorb light at characteristic wavelengths.
- Hemes in the 3 classes of cytochrome (a, b, c) differ slightly in substituents on the porphyrin ring system.
- Hemes a & a₃ are often referred to as cytochromes aa₃.
- Cytochrome c is a small, water-soluble protein with a single heme group.
- The heme iron can undergo an electron transition between ferric and ferrous states:



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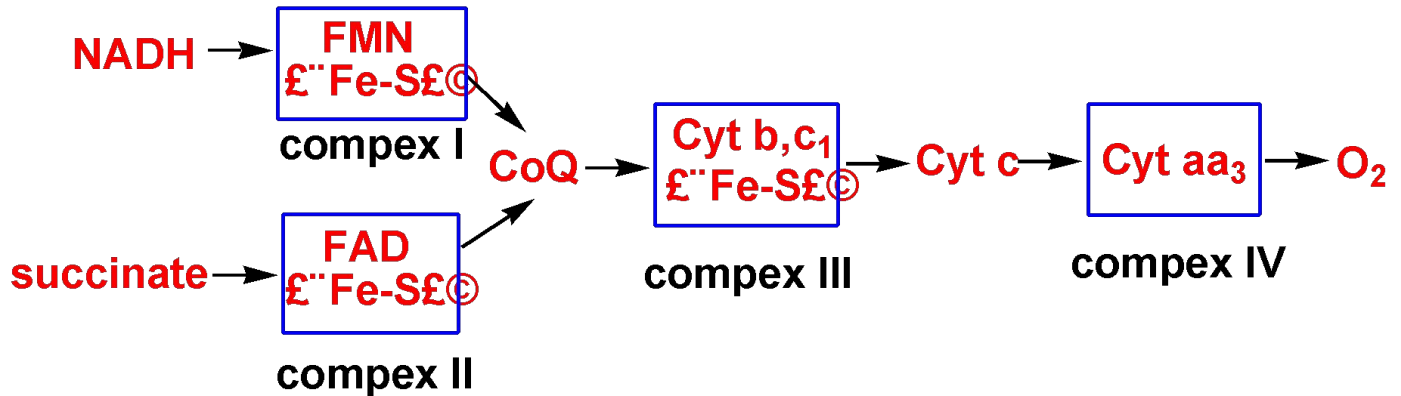


ORGANISATION OF ELECTRON TRANSPORT CHAIN:



(Organisation of electron transport chain and route-map of electron flow through ETC)

Order of Respiratory Chain Complexes

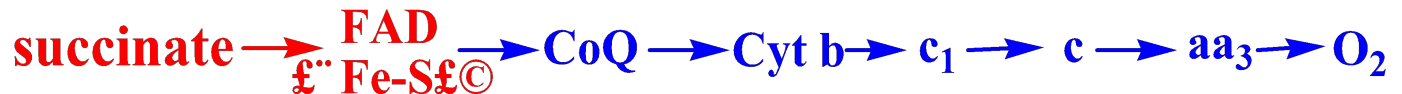


Types of Respiratory Chains

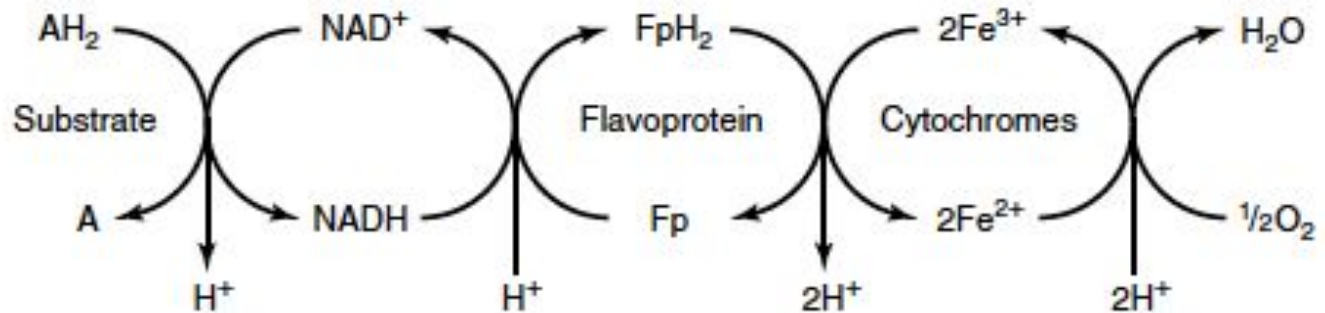
1. NADH respiratory chain :



2. Succinate respiratory chain:

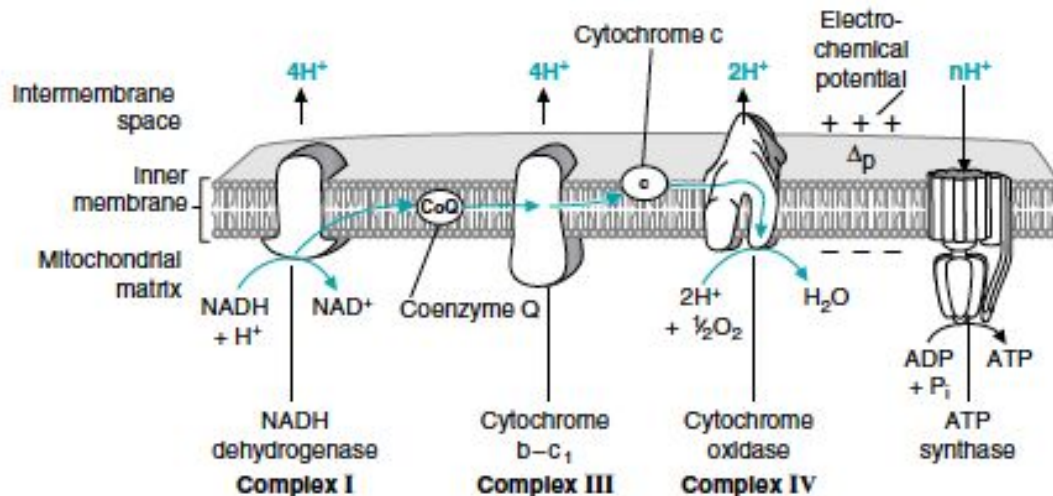


Transport of reducing equivalents through the respiratory chain:

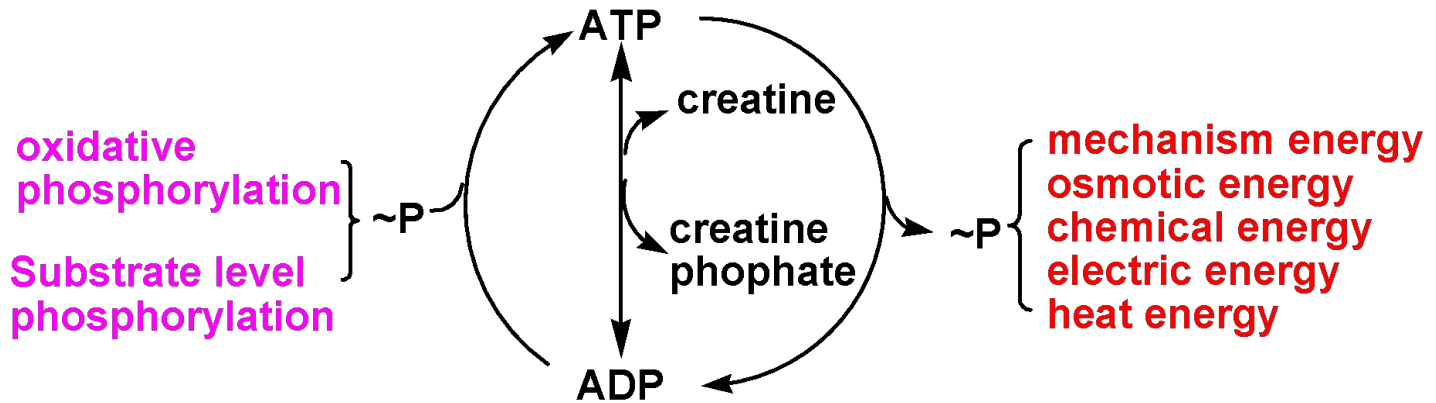


Oxidative phosphorylation:

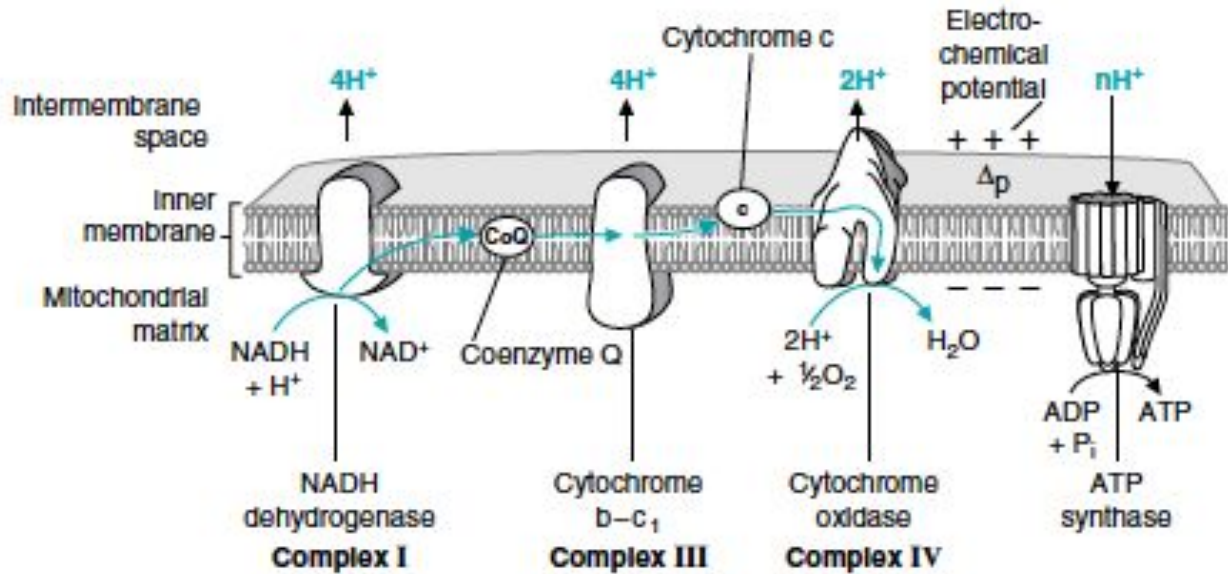
- The phosphorylation of ADP to ATP coupled to electron transfer from a substrate to molecular oxygen.
- Takes place in mitochondria
- Major source of ATP in aerobic organisms.



Production and application of ATP



Oxidative phosphorylation



- Blue arrows show the path of electron transport from NADH to O₂.
- As electrons pass through the chain, protons are pumped from the mitochondrial matrix to the intermembrane space.
- Thereby establishing an electrochemical potential gradient across the inner mitochondrial membrane that drives protons into the matrix through a pore in ATP synthase, which uses the energy to form ATP from ADP and P_i.

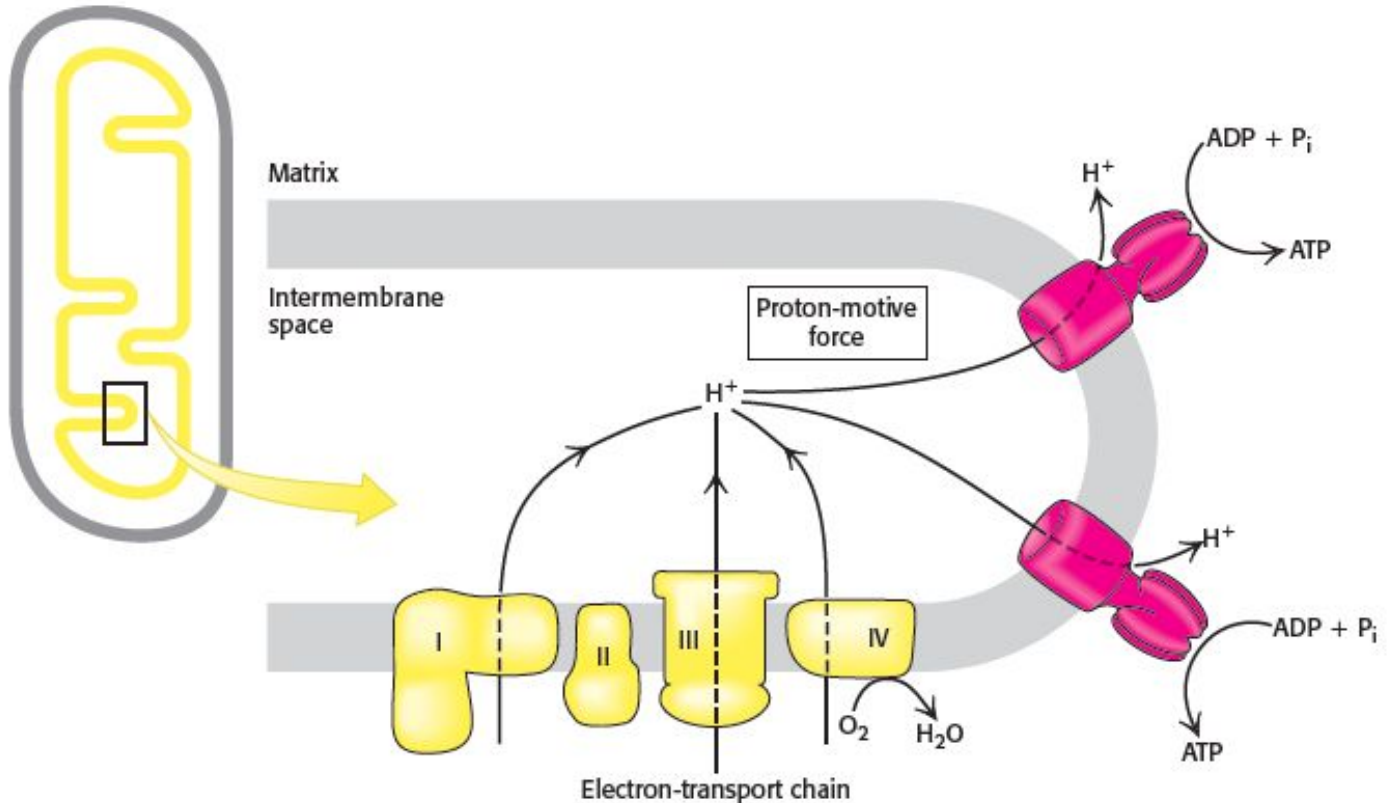
Functional stages in the respiratory chain:

1. H_2 is abstracted from $\text{NADH}+\text{H}^+$ and from FADH_2
2. The electrons obtained with the hydrogen are passed down a cascade of carrier molecules located in complexes I–IV, then transferred to O_2
3. Powered by electron transport, complexes I, III, and IV expel protons across the inner mitochondrial membrane
4. The expelled protons reenter the mitochondrion through ATP synthase, driving ATP synthesis

Why do we need Oxidative Phosphorylation?

- A sedentary male of 70 kg (154 lb) requires ~ 8400 kJ (2000 kcal)/day - equals to 83 kg of ATP
- However, human beings possess only about 250g of ATP (<1% of daily requirement)
- The disparity between the amount of ATP that we have and the amount that we require is solved by recycling spent ATP back to usable ATP
- Each ATP molecule is recycled from ADP approximately 300 times per day
- This recycling takes place primarily through oxidative phosphorylation, in which ATP is formed as a result of the transfer of electrons from NADH or FADH_2 to O_2 by a series of electron carriers

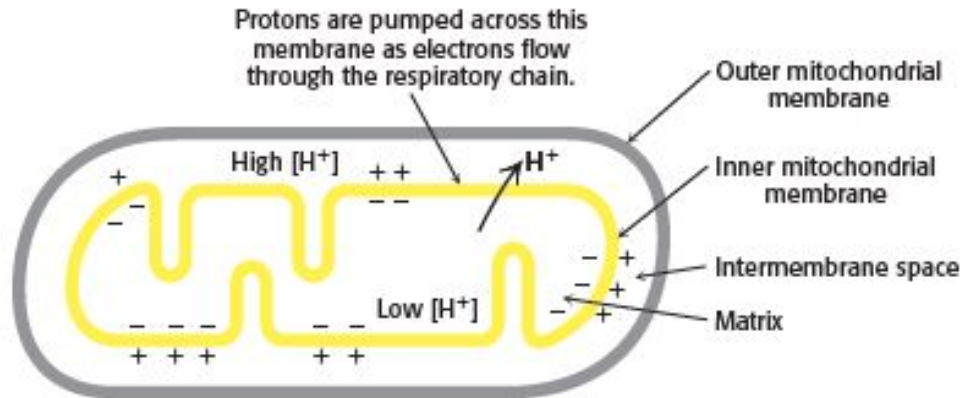
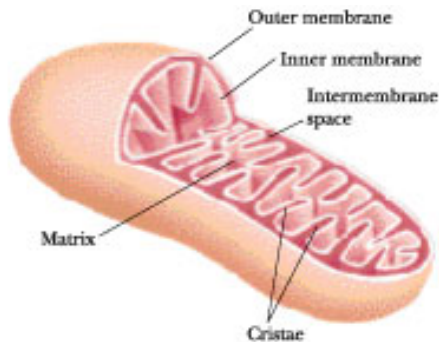
An overview of oxidative phosphorylation

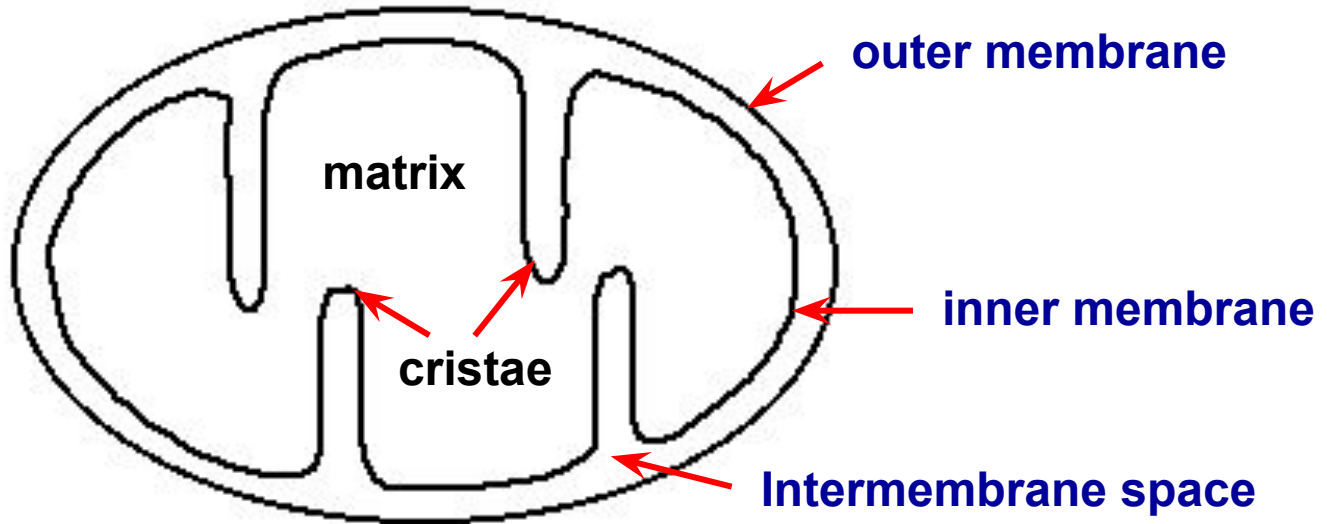


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Chemiosmotic hypothesis:

- According to this concept, a proton concentration gradient serves as the energy reservoir for driving ATP formation
- Originally formulated by Peter Mitchell in the early 1960s.
- Electron transfer through the respiratory chain leads to the pumping of protons from the matrix to the cytoplasmic side of the inner mitochondrial membrane.



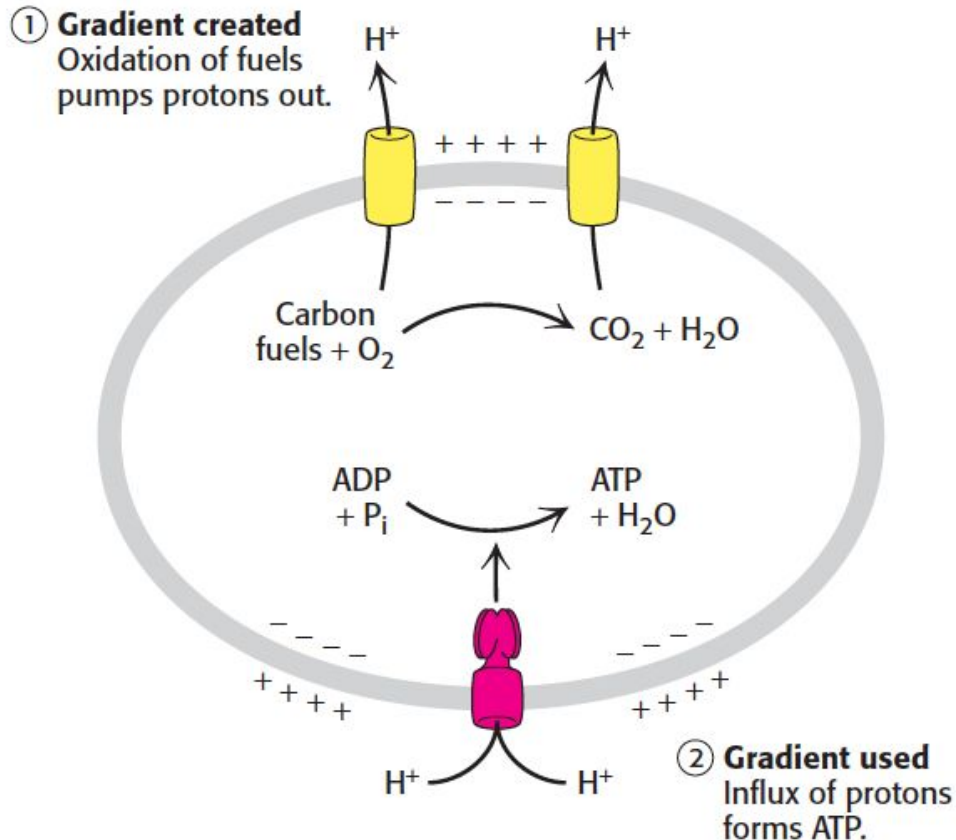


- ♦ Respiratory chain is in cristae of the inner membrane.
- ♦ Spontaneous electron flow through each of complexes I, III, & IV is coupled to H^+ ejection from the matrix.

Chemiosmotic hypothesis (Cont' d):

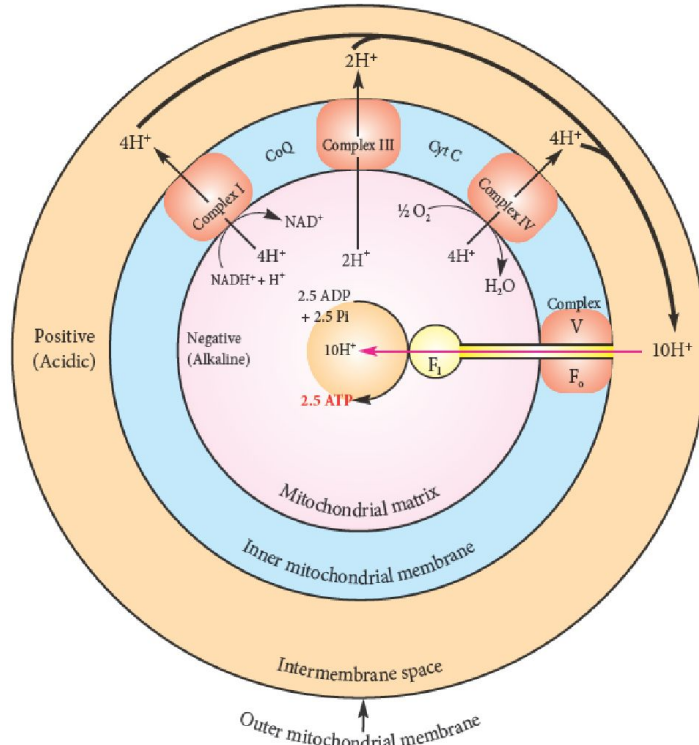
- The pH gradient and membrane potential constitute a proton-motive force that is used to drive ATP synthesis.
- The electrochemical gradient or proton gradient thus developed, acts as the power that drives the mechanism operating the synthesis of ATP.
- The essence of chemiosmotic theory is, the free energy generated from the electron flow through ETC is conserved by pumping protons out of the matrix to create an electrochemical proton gradient across the inner mitochondrial membrane.
 - The electrochemical potential of this proton gradient is used to synthesise ATP.

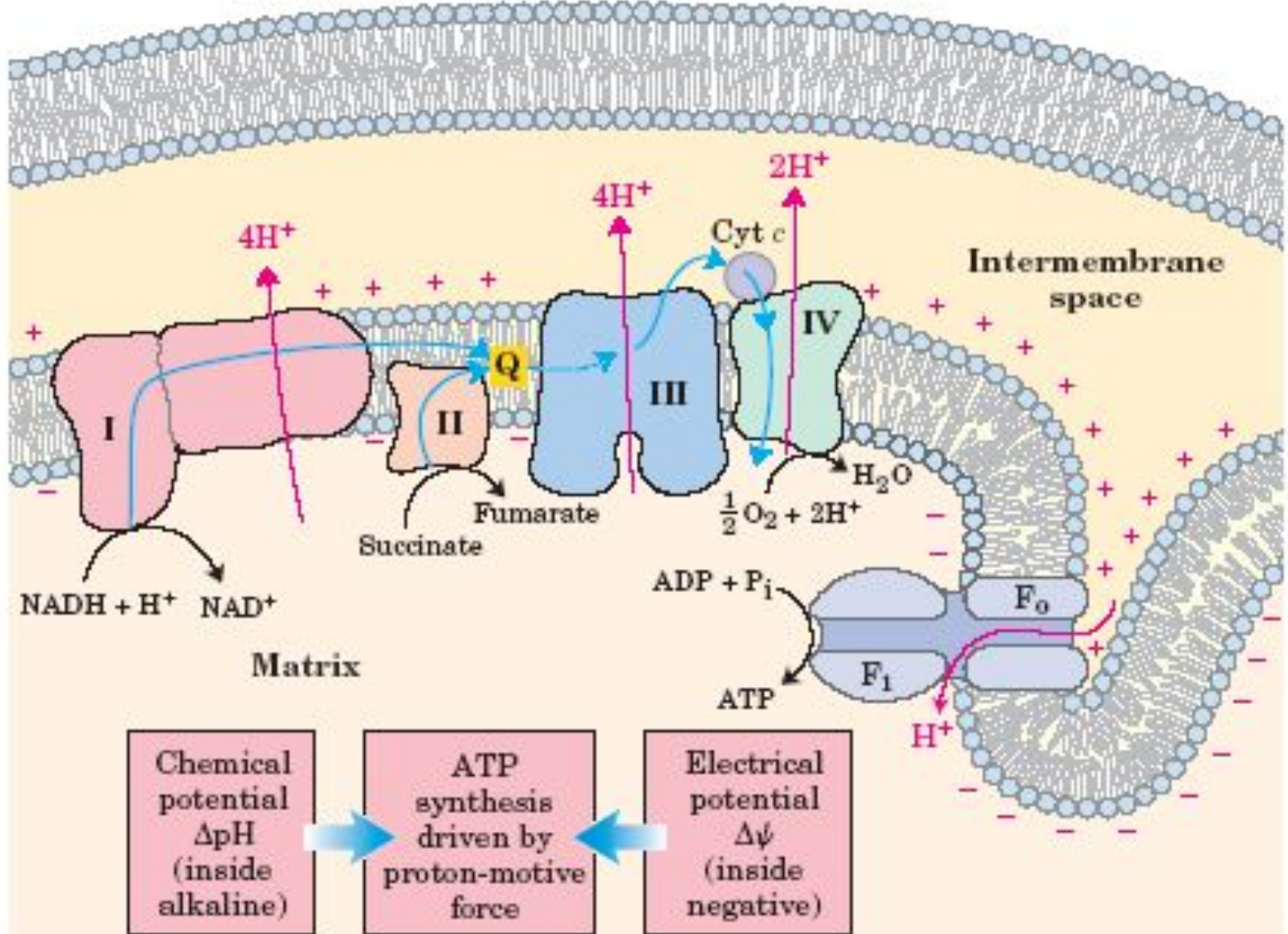
Proton gradients

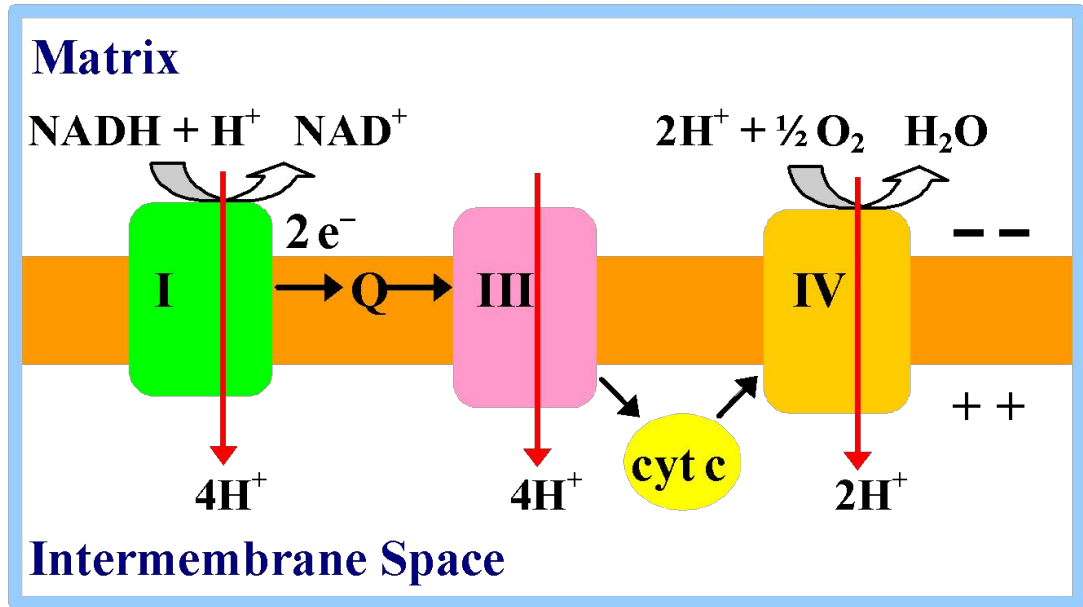


CHEMIOSMOTIC THEORY

- Diagram explaining the chemiosmotic theory of oxidative phosphorylation.

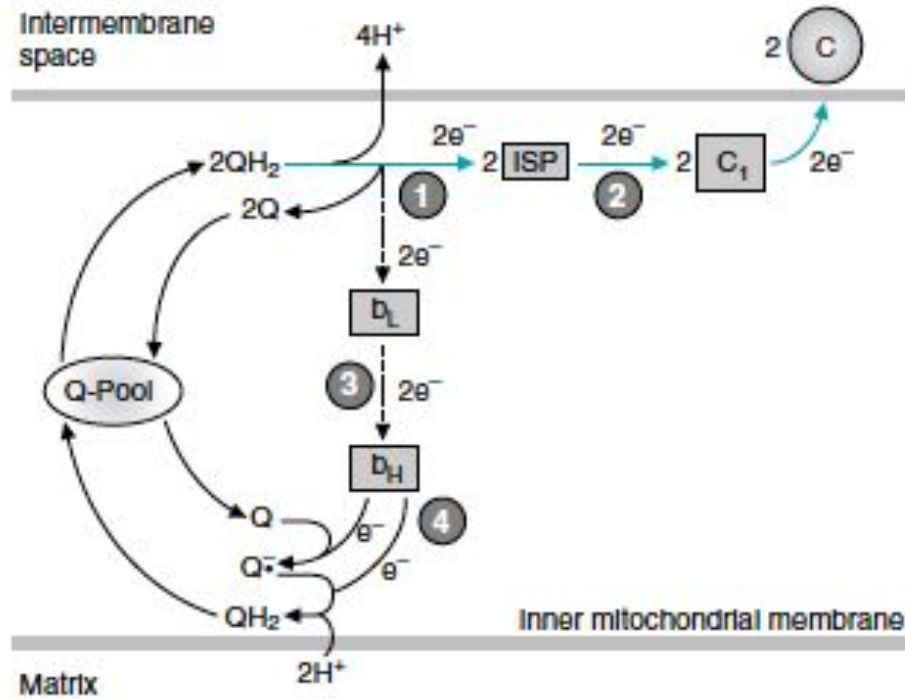






- A total of **10 H^+** are ejected from the mitochondrial matrix **per $2e^-$** transferred from NADH to oxygen via the respiratory chain.

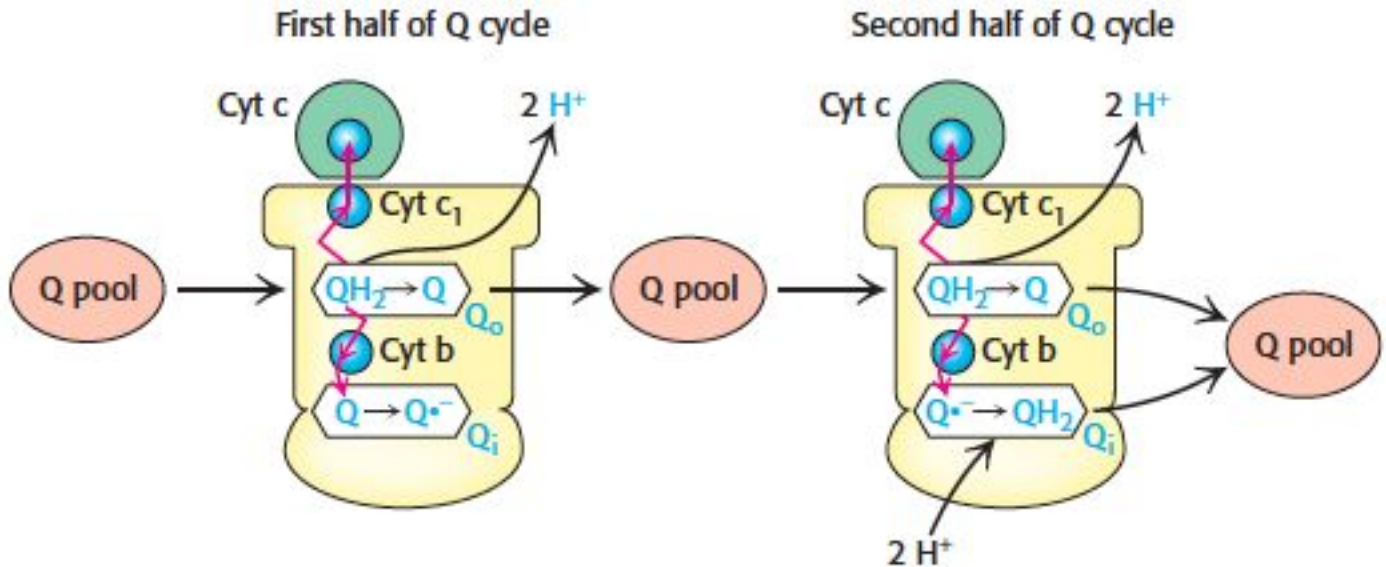
Pumping of Protons: (The proton motive Q cycle for the b-c1 complex)



ISP, FeS center protein; Q, Coenzyme Q; Q^{•-}, CoQ semiquinone; QH₂, CoQ hydroquinone

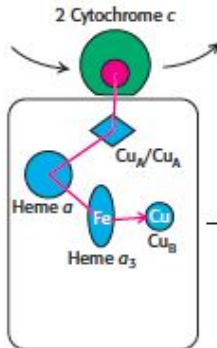
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The Q cycle

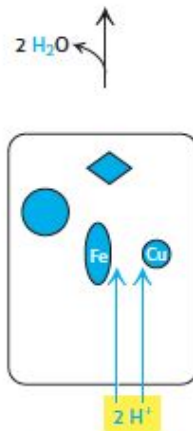
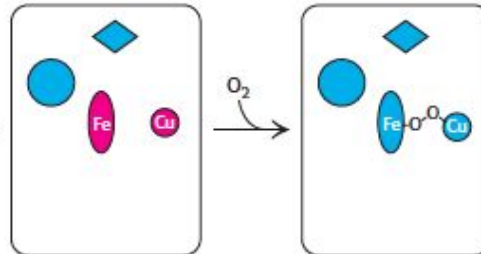


The cytochrome c oxidase mechanism

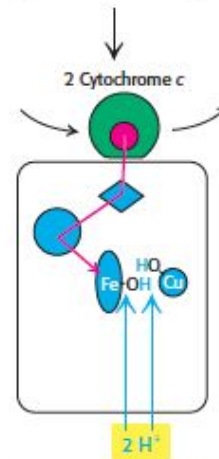
1. Two molecules of cytochrome c sequentially transfer electrons to reduce Cu_B and heme a_3 .



2. Reduced Cu_B and Fe in heme a_3 bind O_2 , which forms a peroxide bridge.



4. The addition of two more protons leads to the release of water.

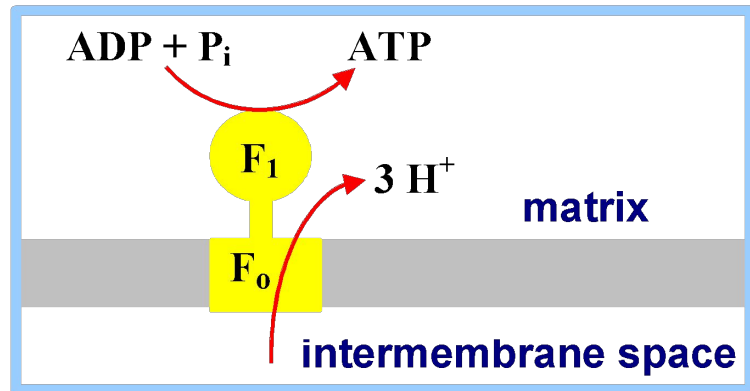
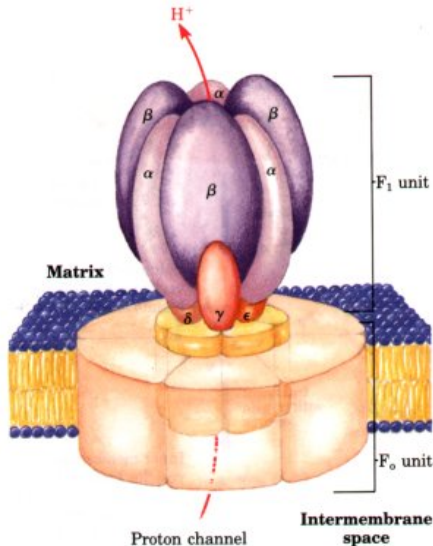


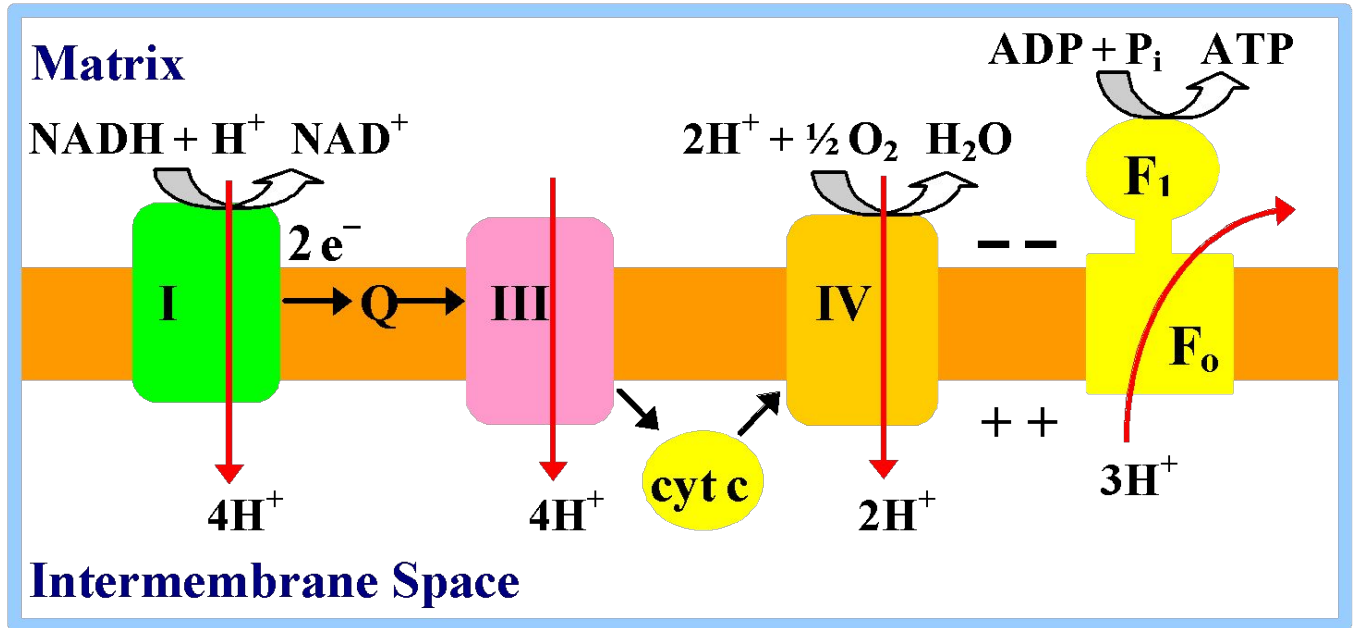
3. The addition of two more electrons and two more protons cleaves the peroxide bridge.

ATP synthase (complex V):

ATP synthase, embedded in cristae of the inner mitochondrial membrane, includes:

- ◆ F_1 catalytic subunit, made of 5 kinds of polypeptides $\alpha_3\beta_3\gamma\delta\epsilon$.
- ◆ F_0 complex of integral membrane proteins mediates proton transport.





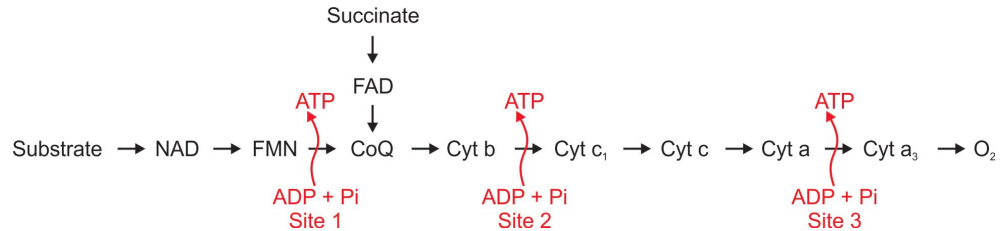
P/O ratio:

- P/O ratio is the number of inorganic phosphates incorporated into ATP per oxygen atom consumed.
(number of ATP / 2H)
- P/O ratio of some substrate:

substrates	components	P/O ratio	Number of ATP
β-hydroxybutyrate	$\text{NAD}^+ \rightarrow \text{O}_2$	2.4~2.8	3
Succinate	$\text{FAD} \rightarrow \text{O}_2$	1.7	2
Vit C	$\text{Cyt c} \rightarrow \text{O}_2$	0.88	1
Cyt c	$\text{Cyt aa}_3 \rightarrow \text{O}_2$	0.61~0.68	1

Coupled sites in ETC:

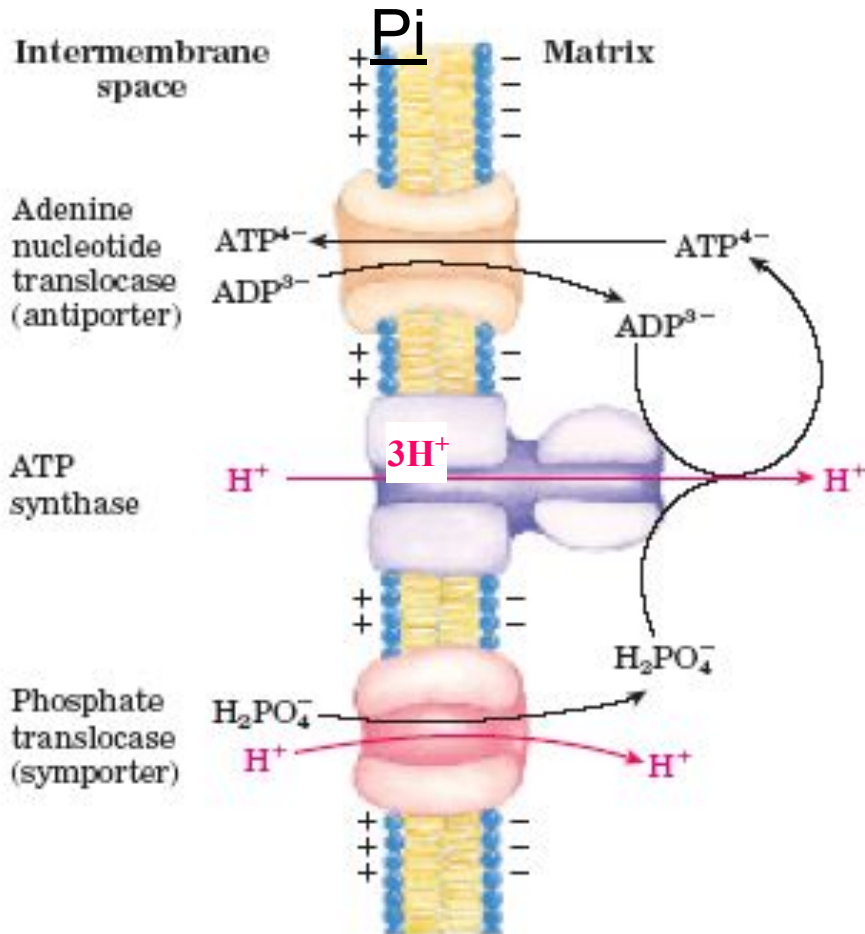
- ① $\text{NADH} \rightarrow \text{CoQ}$
- ② $\text{CoQ} \rightarrow \text{Cyt c}$
- ③ $\text{Cyt aa}_3 \rightarrow \text{O}_2$



1. NADH ETC has 3 coupled site,
P/O ratio =3,
3mol ATP / 2H.

2. Succinate ETC has 2 coupled site,
P/O ratio =2.

Transport of ATP, ADP, &



Summary

- 2.5 ~P bonds synthesized during oxidation of NADH produced in the matrix. (10 H^+ pumped; 4 H^+ used up per ATP).
- 1.5 ~P bonds synthesized during oxidation of FADH_2 produced in the matrix.

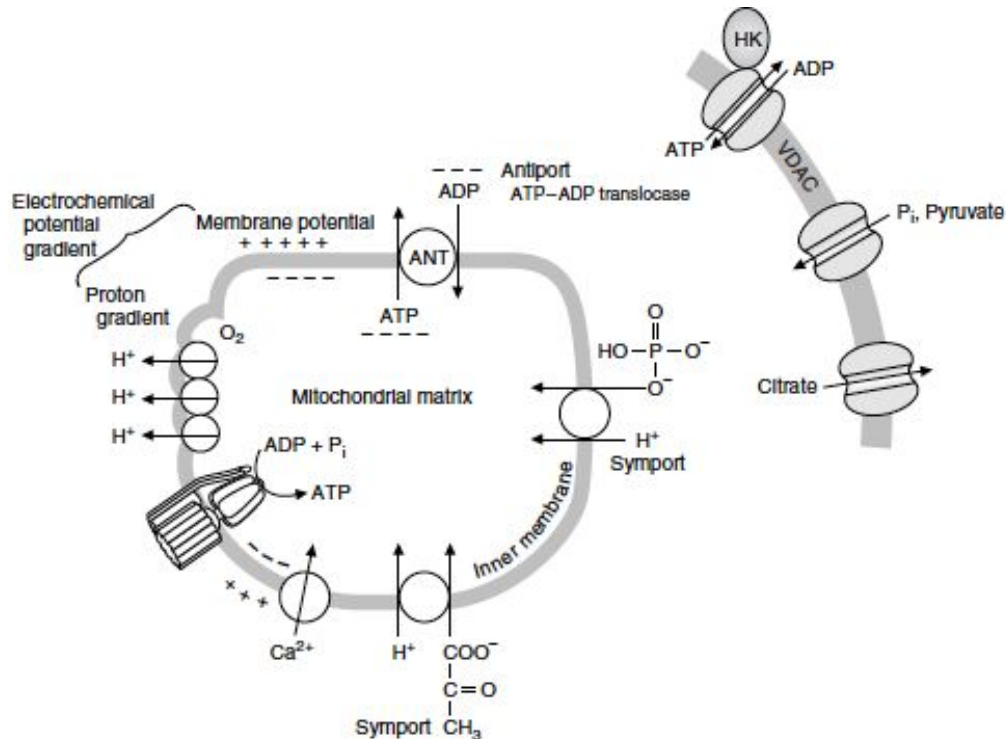
A case study: Luft syndrome

- If the mitochondria of a patient oxidize NADH irrespective of whether ADP is present □ such a defect is called Luft syndrome.
- the P:O ratio for oxidative phosphorylation by these mitochondria is less than normal.
- In this case:
 - Extensive uncoupling of mitochondrial respiration due to leakage of protons.
 - Incapable of performing prolonged physical work.
 - Basal metabolic rate would be more than twice normal, but thyroid function would be normal.
 - A muscle biopsy would show mitochondria were highly variable and atypical in structure.
 - Biochemical studies would reveal that oxidation and phosphorylation are not tightly coupled in these mitochondria.
 - In these patient, much of the energy of fuel molecules would be converted into heat rather than ATP.

Shuttle Systems

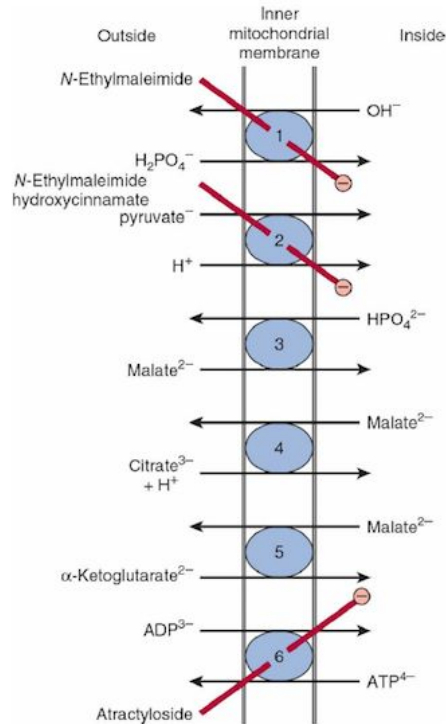
- Oxidation of Extramitochondrial NADH Is Mediated by Substrate Shuttles
- The inner membrane of mitochondria is quite impermeable to some molecules and ions.
- Permeable to:
 - Pyruvate, succinate, α -ketoglutarate, malate, citrate, Glutamate etc.
- Impermeable to:
 - H^+ , NADH, NADPH, OAA etc.

Transport of compounds across the inner and outer mitochondrial membranes.



ANT, adenine nucleotide translocase; VDAC, voltage-dependent anion channels

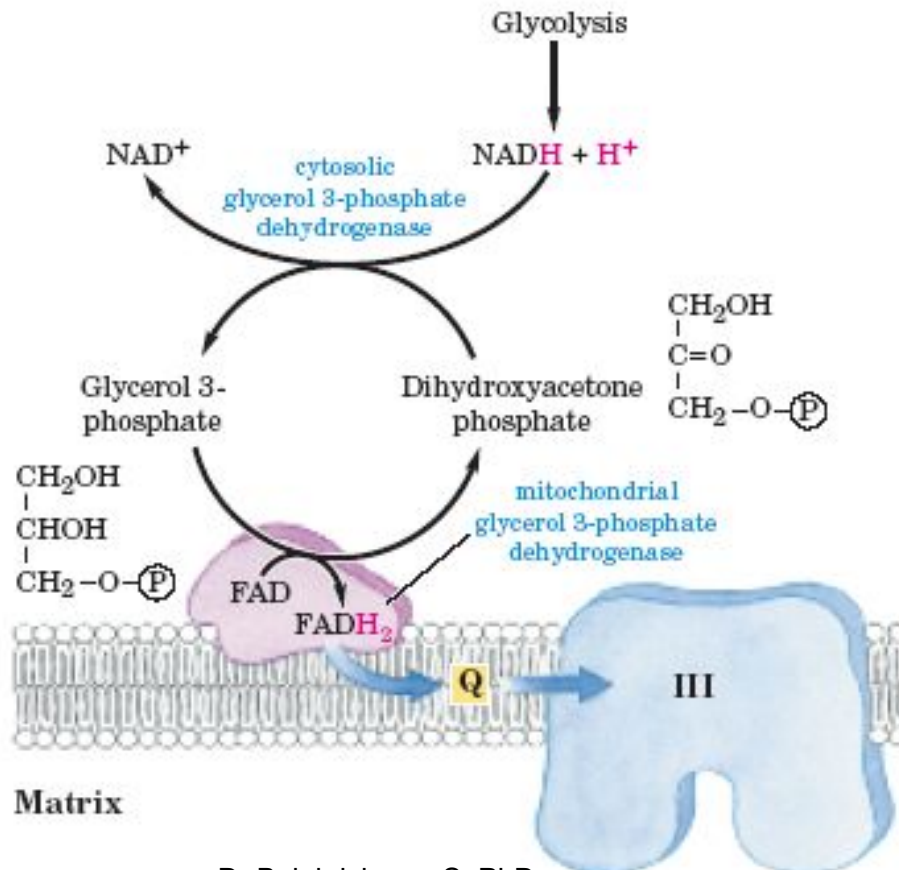
Transporter systems in the inner mitochondrial membrane



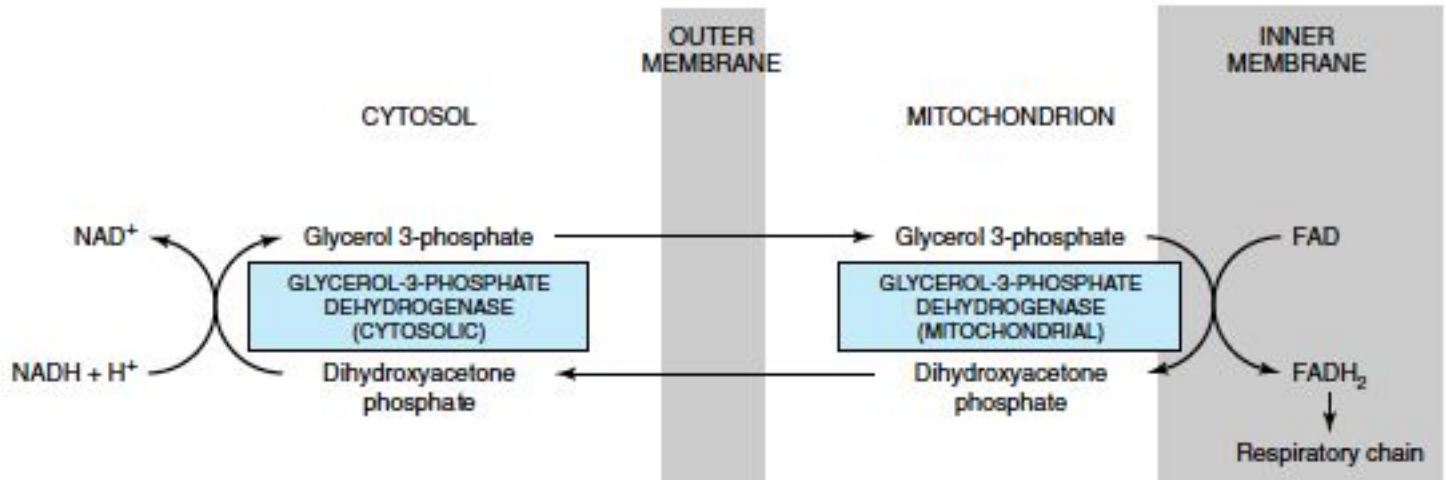
1 - Phosphate transporter, 2 - pyruvate symport, 3 - dicarboxylate transporter, 4 - tricarboxylate transporter, 5 – α -ketoglutarate transporter, 6 - adenine nucleotide transporter.

α - Glycerophosphate shuttle system

(Brain, skeletal muscle but deficient in heart muscle)

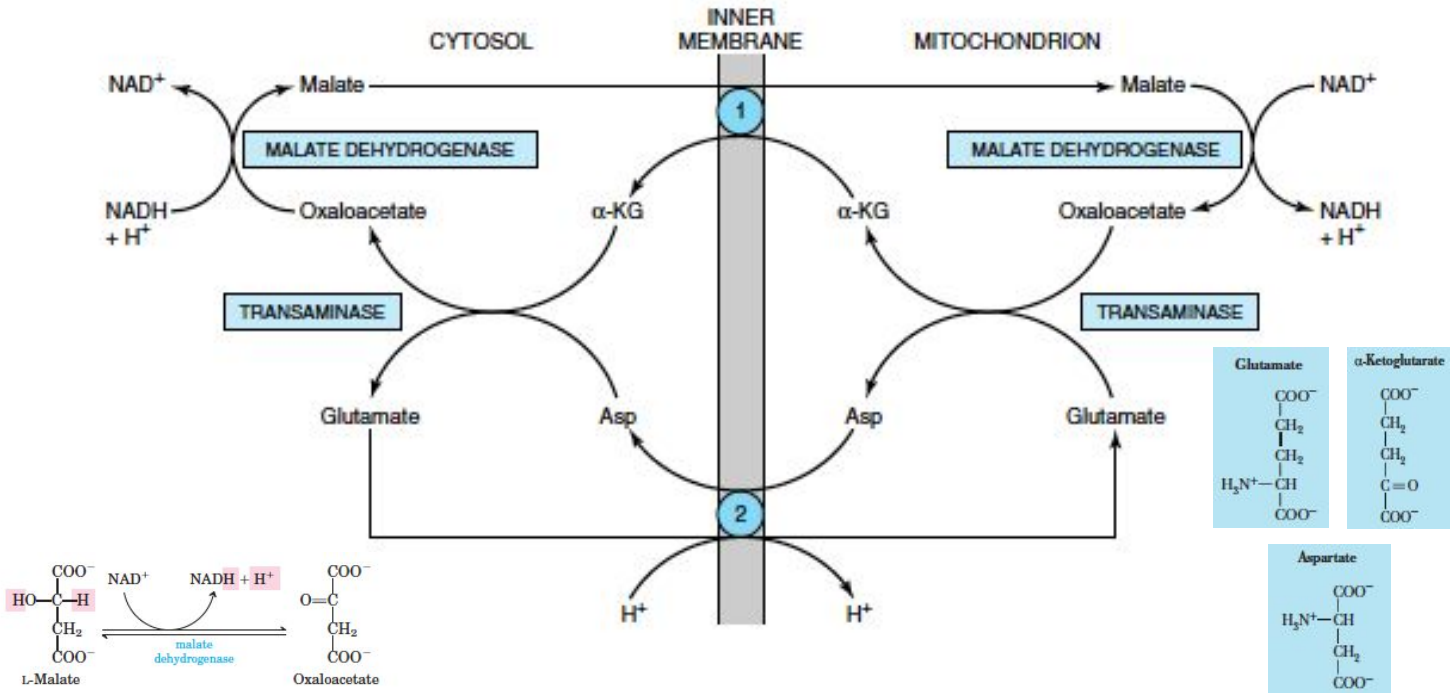


Glycerophosphate shuttle for transfer of reducing equivalents from the cytosol into the mitochondrion:



In Muscle

Malate shuttle for transfer of reducing equivalents from the cytosol into the mitochondrion



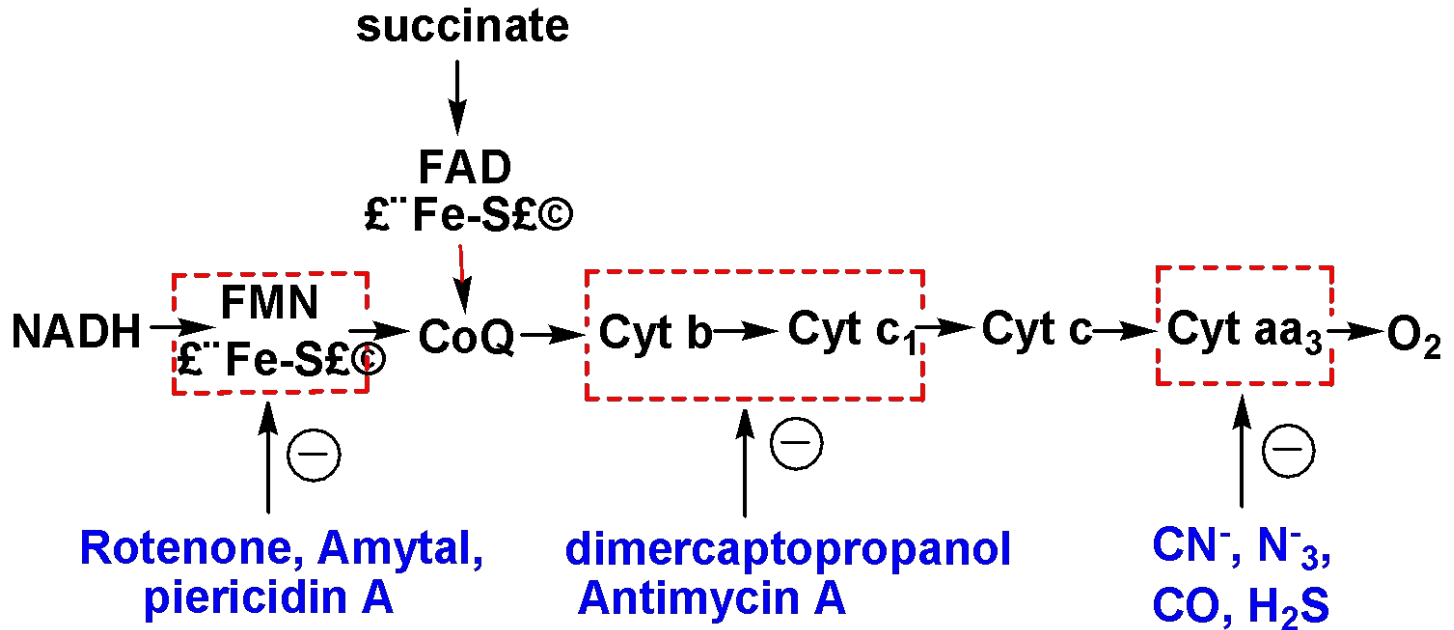
Malate-aspartate shuttle (heart, liver)

REGULATION OF ETC & OXIDATIVE PHOSPHORYLATION

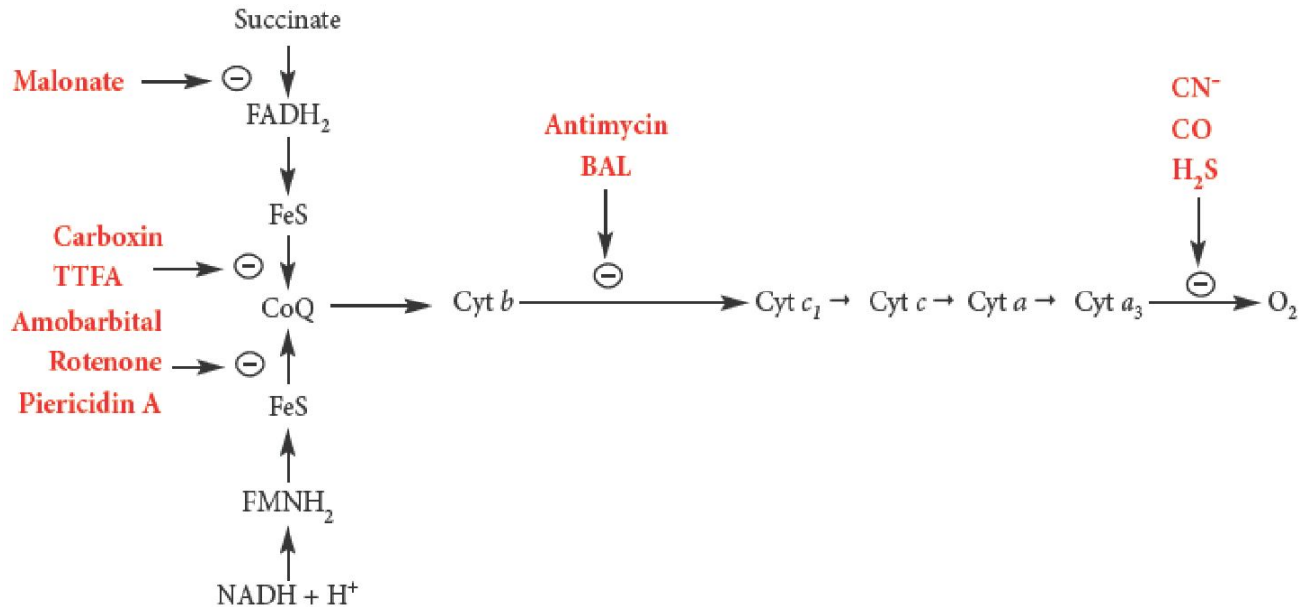
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1. The factors affecting ETC:

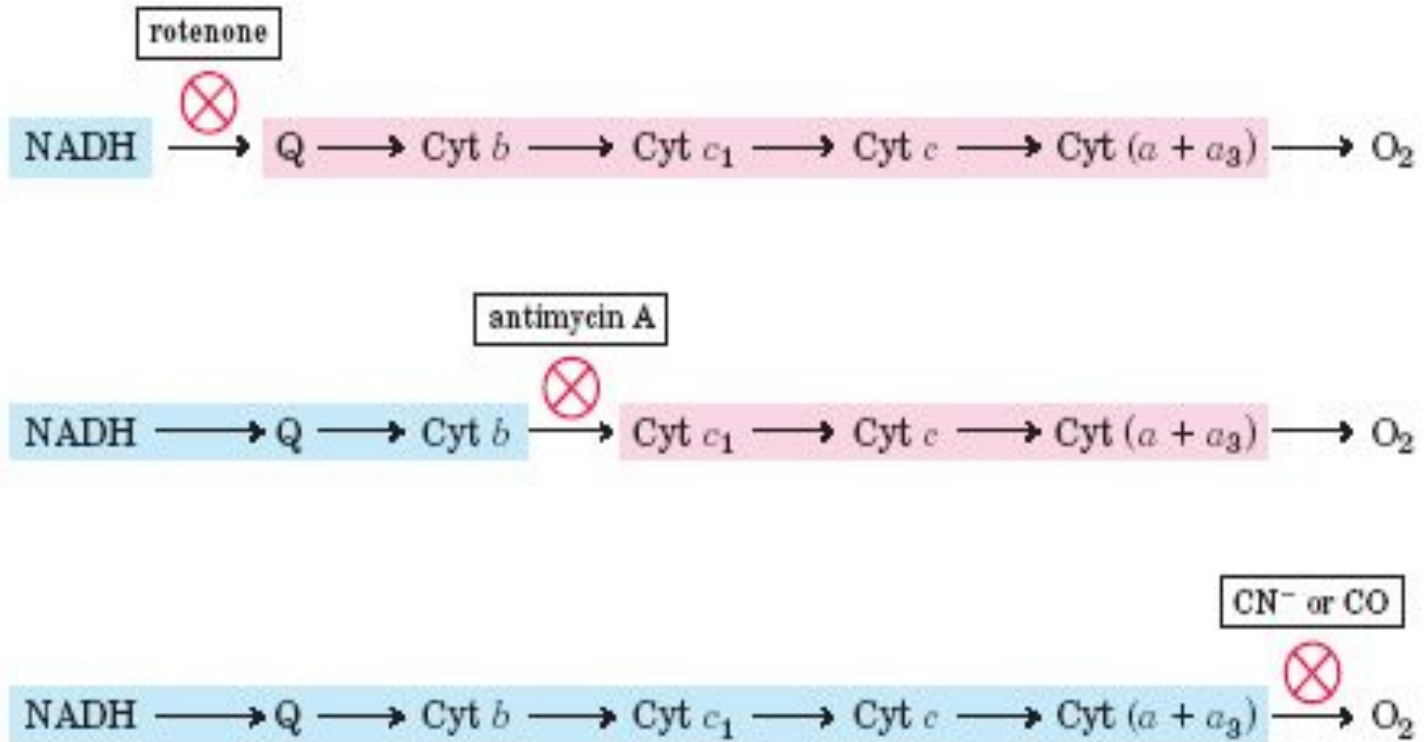
A. Inhibitors of ETC:



INHIBITORS OF THE ELECTRON TRANSPORT CHAIN



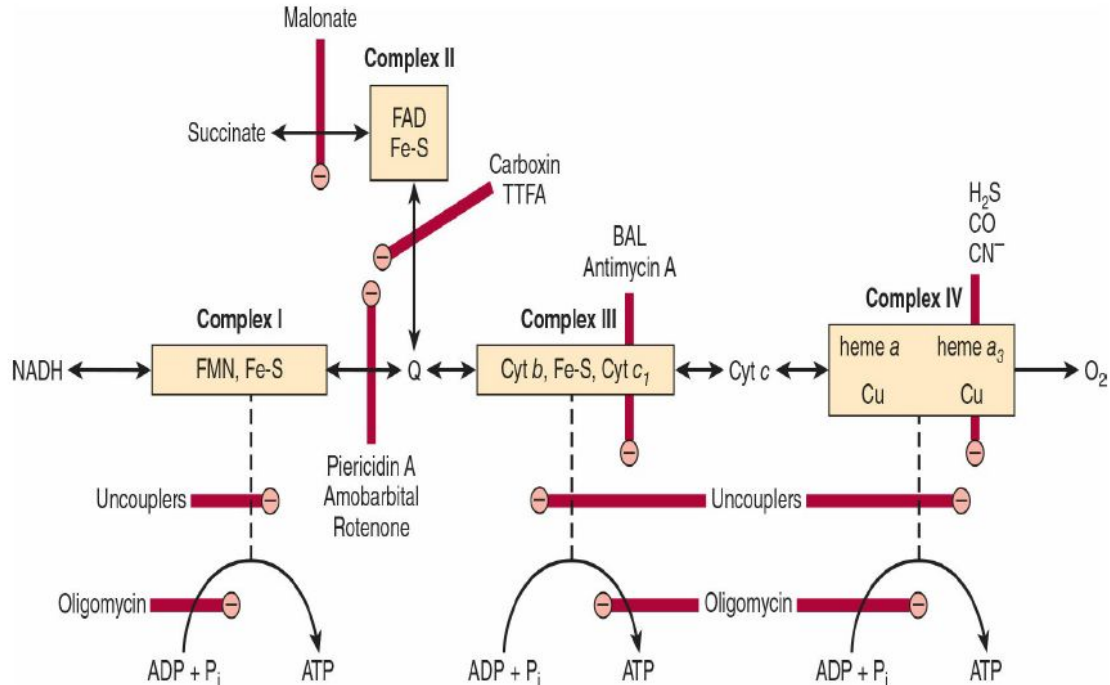
Site specific inhibitors of ETC



Cyanide antidote - Nitrite

- Cyanide can be lethal because it binds to the ferric form of cytochrome oxidase and thereby inhibits oxidative phosphorylation.
- The immediate administration of nitrite is a highly effective treatment for cyanide poisoning.
- Nitrite converts ferrohemoglobin into ferrihemoglobin, which also binds cyanide.
 - Thus, ferrihemoglobin competes with cytochrome oxidase for cyanide.
 - This competition is therapeutically effective because the amount of ferrihemoglobin that can be formed without impairing oxygen transport is much greater than the amount of cytochrome oxidase.

Sites of inhibition of the respiratory chain by specific drugs, chemicals, and antibiotics.

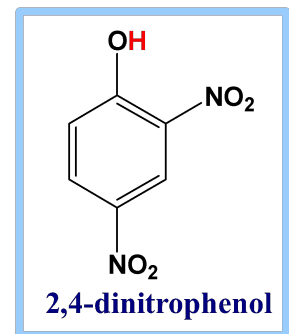


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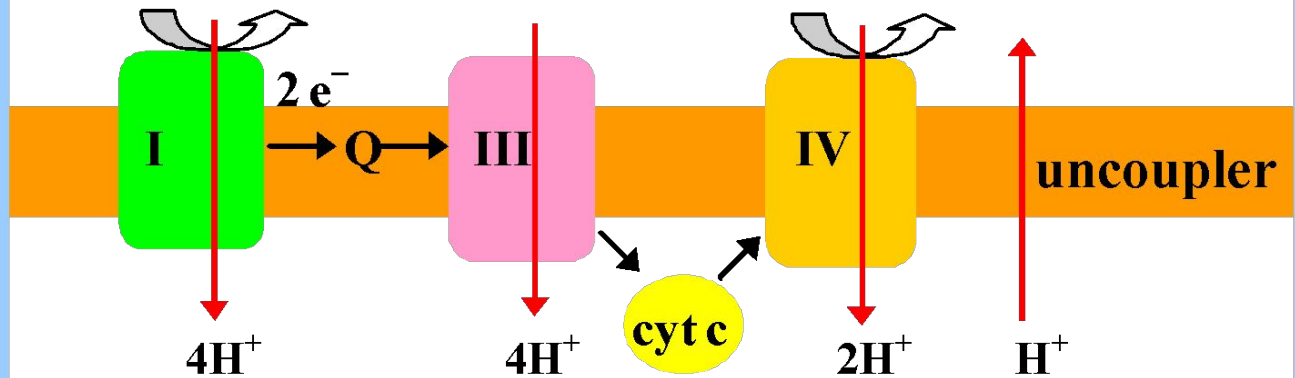
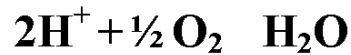
BAL = British anti-Lewisite; Dimercaprol
TTFA = Thenoyltrifluoroacetone

B. Uncoupling agent

- A compound that disrupts the usual tight coupling between electron transport and phosphorylation of ADP.
- The action of uncouplers is to dissociate oxidation in the respiratory chain from phosphorylation.
- They dissipate the electrochemical gradient across the membrane without generating ATP
- Uncoupling reagents dissolve in the membrane and function as carriers for H^+ .
- Examples:
 - 2,4-Dinitrophenol (DNP)
 - Dicuomarol (an anticoagulant).
 - Salicylate, a metabolite of aspirin.



Matrix

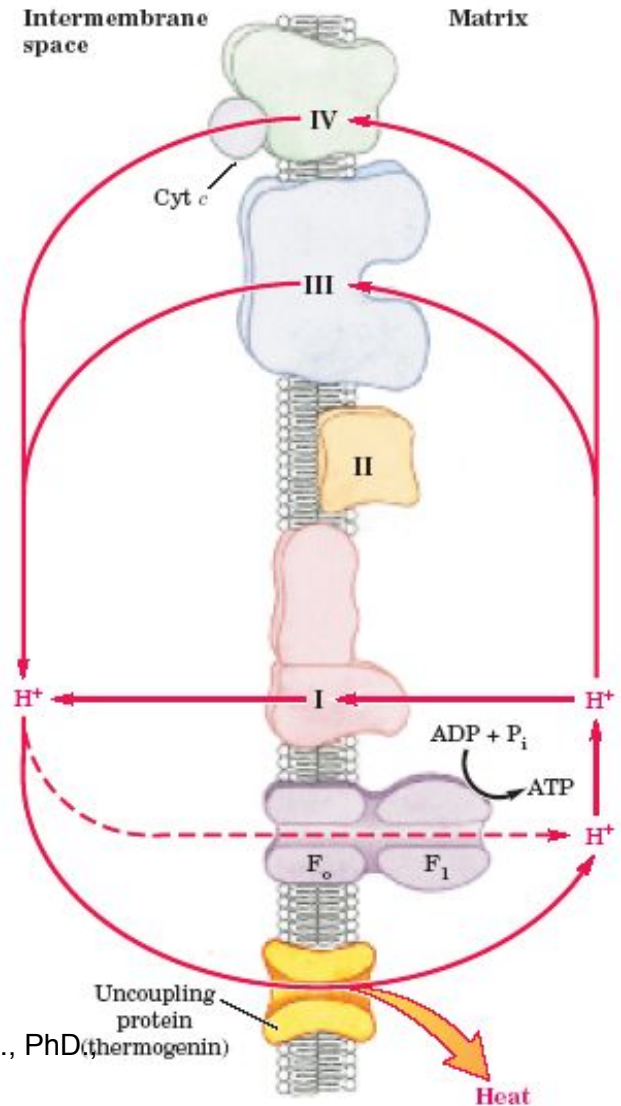


Intermembrane Space

- Uncouplers block oxidative phosphorylation by dissipating the H⁺ electrochemical gradient.

Physiological uncouplers:

- Endogenous Uncouplers Enable Organisms To Generate Heat.
- Example: Thermogenin & Thyroxine



Dr Balakrishnan. S., PhD(thermogenin)

C. Inhibiter of oxidative phosphorylation

- A compound that Inhibits electron transport and phosphorylation of ADP.
- Example: Oligomycin

2. ADP

$[ADP] \uparrow \rightarrow \text{Oxidative phosphorylation} \uparrow$

$[ADP] \downarrow \rightarrow \text{Oxidative phosphorylation} \downarrow$

3. Thyroxine

- $\text{Na}^+, \text{K}^+ \text{-ATPase} \rightarrow \text{ATP catabolism} \rightarrow \text{ADP} \uparrow \rightarrow \text{Oxidative phosphorylation} \uparrow \rightarrow \text{formation of ATP} \uparrow$.
- Formation and catabolism of $\text{ATP} \uparrow$, consuming of $\text{O}_2 \uparrow$, Generation of $\text{Heat} \uparrow$.

4. mtDNA mutation

- Free radical → mutation of DNA → inhibition of Oxidative phosphorylation → energy↓

Deficiencies of electron transport

- Impaired electron transfer through any complex can have pathologic consequences.
 - Fatigue can result from iron-deficiency anemia , which decreases Fe for Fe-S centers and cytochromes.
- Oxphos diseases are caused by mutations in nuclear DNA or mtDNA that decrease mitochondrial capacity for oxidative phosphorylation.
 - Mitochondrial DNA (mtDNA), which is maternally inherited, encodes some of the subunits of the electron transport chain complexes and ATP synthase.

DISORDERS ASSOCIATED WITH ETC AND OXIDATIVE PHOSPHORYLATION

- Alzheimer's and Parkinson's diseases
- Leber's hereditary optic neuropathy
- MELAS (Mitochondrial Encephalopathy, Lactic Acidosis and Stroke)

MELAS (Mitochondrial Encephalopathy, Lactic Acidosis and Stroke):

- What is MELAS?
 - Mitochondrial = Mutation in the mitochondria;
 - Encephalo- = Occurs in the head;
 - Myopathy = General term for muscle disease;
 - Lactic Acidosis = The blood gets too acidic;
 - Stroke-like Episodes = Brain-related symptoms of bleeding or blockage
- Most common maternally inherited mitochondrial disease.
- At least 6 different point mutations associated with MELAS.
 - 80% - A to G point mutation in the tRNA (Leu) gene at bp 3243.
 - 7.5% - T to C point mutation in the tRNA (Leu) gene at bp 3271.

MELAS (Mitochondrial Encephalopathy, Lactic Acidosis and Stroke):

- Biochemical Mechanism:
 - Abnormal mitochondria do not metabolize pyruvate.
 - $\square$ pyruvate $\square$ Lactic acid (accumulates in blood and other fluids).
 - Large clumps of abnormal mitochondria form in the walls of small arteries and capillaries in the brain and muscles
 - Clinical Features: Strokes, myopathy, muscle twitching, dementia, and deafness

Read more...



THANK YOU...

Dr Balakrishnan. S., PhD.,